COMPREHENSIVE
PEPTIDE GUIDE

Science · Chemistry · Clinical Trials · Applications

A Complete Reference on Therapeutic Peptides · For Educational Purposes Only

50+	30+	12	FDA
Peptides Covered	Clinical Trials	Categories	Approved Listed

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1 What Are Peptides?

A Complete Scientific Primer

Definition & Structure

Peptides are short chains of **amino acids** — the same fundamental building blocks that make up proteins — linked together by covalent peptide bonds (also called amide bonds). By convention, a chain of 2 to 50 amino acids is classified as a peptide, while longer chains become proteins. This distinction matters biologically: the shorter length of peptides allows them to be highly specific in their biological activity, moving rapidly through tissues and targeting precise cellular receptors with extraordinary selectivity.

The structure of a peptide is defined by its **primary sequence** — the specific order of amino acids from its N-terminus (amino end) to its C-terminus (carboxyl end). This sequence determines how the peptide folds in three-dimensional space, which in turn determines which biological receptors it can interact with. A change of even a single amino acid — or a slight modification like cyclization, acetylation, or PEGylation — can dramatically alter potency, selectivity, and half-life.

Amino Acid	Abbreviation	Role in Peptides
Glycine	Gly / G	Simplest AA; adds flexibility to chain
Leucine	Leu / L	Hydrophobic core stability; common in GHS peptides
Lysine	Lys / K	Positive charge; receptor binding; conjugation site
Arginine	Arg / R	Positive charge; DNA binding; signal transduction
Cysteine	Cys / C	Forms disulfide bonds — critical for cyclic peptides
D-Amino acids	D-Phe, D-Ala...	Mirror-image AAs used in synthetic peptides; resist enzymatic degradation

Key amino acid building blocks and their roles in therapeutic peptides.

Natural vs. Synthetic Peptides

The human body naturally produces hundreds of bioactive peptides that serve as **hormones, neurotransmitters, growth factors, immune regulators, and antimicrobial agents**. Examples include insulin (51 amino acids), oxytocin (9 amino acids), and the enkephalins (pain-modulating pentapeptides). These endogenous peptides are produced in specific tissues, released in response to physiological signals, and act at precise target sites throughout the body.

Synthetic therapeutic peptides are designed to either **mimic** natural peptides (analogs), **modify** natural sequences for improved properties (half-life, potency, stability), or represent entirely **novel sequences** designed to modulate specific biological targets. Modifications commonly include: D-amino acid substitutions (resisting proteolytic enzymes), cyclization (structural stability), PEGylation (extended half-life via reduced renal clearance), fatty acid conjugation (albumin binding for long action), and C-terminal amidation (reduced renal clearance).

Why Peptides Over Traditional Drugs?

Peptides occupy a unique therapeutic niche between small-molecule drugs and large biologics. They offer: (1) High receptor specificity reducing off-target effects; (2) Natural breakdown into harmless amino acids (no toxic metabolites); (3) Biological activity at nanomolar concentrations; (4) Ability to modulate complex biological pathways that small molecules cannot reach. Their main limitations are oral bioavailability (most require injection) and shorter half-lives than antibodies — both of which chemical engineering increasingly overcomes.

2 How Peptides Work in the Body

Mechanisms & Pharmacology

Receptor Binding & Signal Transduction

The primary mechanism of most therapeutic peptides is **receptor binding**. Each peptide has a specific three-dimensional shape complementary to one or more receptor proteins on cell surfaces. This 'lock and key' interaction (or more accurately, 'induced fit') triggers a cascade of intracellular signaling events. For example, GLP-1 agonists (semaglutide, tirzepatide) bind transmembrane G-protein coupled receptors (GPCRs), activating adenylyl cyclase and increasing intracellular cAMP, which then triggers insulin secretion from pancreatic beta cells.

Key Pharmacokinetic Concepts

Concept	Definition	Peptide Relevance
Half-Life (t½)	Time for plasma concentration to fall by 50%	Most peptides: minutes to hours; modified analogs: days to weeks
Bioavailability	% of dose reaching systemic circulation	SC injection: 70–90%. Oral: usually <5% (proteolysis in gut)
Volume of Distribution	Apparent body volume drug distributes into	Small Vd = stays in plasma; large Vd = tissue penetration
Proteolysis	Enzymatic degradation of peptide bonds	Primary reason peptides need injection; D-AAs and cyclization resist it
Receptor Selectivity	Preference for target receptor vs. others	Ipamorelin (selective) vs. GHRP-6 (non-selective) illustrate clinical impact
BBB Penetration	Ability to cross blood-brain barrier	Small lipophilic peptides (Semax, Selank) penetrate; most large peptides do not

Administration Routes & Bioavailability

The route of administration fundamentally determines a peptide's bioavailability and clinical utility. **Subcutaneous (SC) injection** is the gold standard — a small gauge needle (29–31G insulin syringe) deposits the peptide into subcutaneous fat, where it is gradually absorbed into capillaries. Bioavailability is typically 70–90%. **Intranasal** routes work for small, lipophilic peptides (Semax, Selank, oxytocin) that can cross the nasal mucosa and, uniquely, can bypass the blood-brain barrier via the olfactory nerve pathway. **Oral** peptides face destruction by stomach acid and proteolytic enzymes — only MK-677 (a peptidomimetic small molecule) and specially formulated oral GLP-1s achieve meaningful systemic bioavailability.

3 Peptide Categories at a Glance

Overview of All Major Classes

Category	Primary Purpose	Key Mechanisms	Examples	Evidence
GH Secretagogues	Increase GH/IGF-1; body composition	GHRH / GHS-R1a activation	Ipamorelin, CJC-1295, Sermorelin, MK-677	★★★–★★★★
Healing & Recovery	Tissue repair; wound healing; anti-inflammatory	VEGF/EGF/angiogenesis pathways; actin modulation	BPC-157, TB-500, GHK-Cu	★★★■ (animal)
GLP-1 / Metabolic	Diabetes; obesity; CV risk	GLP-1R / GIP-R agonism	Semaglutide, Tirzepatide, Liraglutide	★★★★★ FDA Approved
Cognitive / Neuro	Brain health; memory; anxiety	BDNF/NGF upregulation; GABA/dopamine modulation	Semax, Selank, Dihexa, Cerebrolysin	★★–★★★★
Sexual Health	Libido; erectile function; desire	Central MC3R/MC4R; hypothalamic activation	PT-141 (Vyleesi), Kisspeptin	★★★★ FDA (Vyleesi)
Immune Modulation	Immune support; infection; autoimmune	T-cell / NK activation; TLR signaling	Thymosin Alpha-1, LL-37, KPV	★★★–★★★★
Anti-Aging	Telomere extension; longevity; mitochondria	Telomerase activation; AMPK; epigenetics	Epitalon, MOTS-c, Humanin	★★■ (early)
Cardiovascular	Heart failure; ischemia; fluid balance	Mitochondrial cardiolipin; natriuretic receptors	SS-31, Nesiritide, ANP analogs	★★★–★★★★
Skin & Cosmetic	Collagen; pigmentation; anti-wrinkle	Melanocortin; SNARE inhibition; fibroblast activation	GHK-Cu, MTII, Snap-8, Matrixyl	★★–★★★
Pain Management	Severe chronic pain	N-type Ca^{2}■ channel blockade	Ziconotide (Prialt)	★★★★★ FDA Approved
Hormonal / Endocrine	Cancer; fertility; obstetric	GnRH / somatostatin receptor modulation	Leuprolide, Octreotide, Oxytocin	★★★★★ FDA Approved

4 Complete Peptide Directory

Chemical Composition · Mechanisms · Trials · Safety

4.1 GH Secretagogue

Peptides in the GH Secretagogue category

Sermorelin | GH Secretagogue

FORMULA	C149H246N44O42S (29-amino acid peptide; GHRH analog)
MOL. WEIGHT	3,358 Da
SEQUENCE	Tyr-Ala-Asp-Ala-Ile-Phe-Thr-Asn-Ser-Tyr-Arg-Lys-Val-Leu-Gly-Gln-Leu-Ser-Ala-Arg-Lys-Leu-Leu-Gln-Asp-Ile-Met-Ser-Arg-NH2
MECHANISM	Binds GHRH receptors on pituitary somatotrophs, stimulating pulsatile endogenous GH secretion without suppressing the hypothalamic-pituitary axis. Works within natural physiological feedback loops.
CLINICAL USES	Adult GH deficiency, anti-aging, body composition optimization (lean mass, fat reduction), sleep quality improvement, recovery from injury, bone density support.
TYPICAL DOSE	200–500 mcg subcutaneous, administered at bedtime (coincides with natural GH pulse)
HALF-LIFE	10–20 minutes
ROUTE	Subcutaneous injection
SIDE EFFECTS	Injection site reactions, transient flushing, water retention, headaches. Rare: pituitary desensitization with chronic overuse.
CLINICAL TRIALS	FDA approved as Geref for pediatric GH deficiency (now discontinued). Multiple Phase II adult studies. Widely studied in anti-aging medicine; shown to restore youthful GH pulsatility in adults.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Active neoplasms, uncontrolled hypothyroidism, pituitary tumor.

Ipamorelin | GH Secretagogue

FORMULA	C38H49N9O5 (pentapeptide; Aib-His-D-2-Nal-D-Phe-Lys-NH2)
MOL. WEIGHT	711.9 Da
SEQUENCE	Aib-His-D-2-Nal-D-Phe-Lys-NH2
MECHANISM	Selectively activates ghrelin/GHS-R1a receptors on pituitary, triggering GH release. Unique for its high selectivity — does not significantly elevate cortisol, ACTH, or prolactin unlike other GHRPs. This makes it the safest GHRP.
CLINICAL USES	Body recomposition (lean muscle gain, fat loss), sleep quality, GI motility disorders, post-surgical recovery, anti-aging.
TYPICAL DOSE	200–300 mcg subcutaneous, 1–3× daily or at bedtime
HALF-LIFE	~2 hours
ROUTE	Subcutaneous injection
SIDE EFFECTS	Generally very well tolerated. Mild water retention, transient hunger, occasional headache. Minimal cortisol/prolactin elevation.
CLINICAL TRIALS	Extensive Phase I/II data. Studied for post-operative ileus and GI motility (Novo Nordisk). Numerous published pharmacokinetic and GH-releasing studies. FDA banned from compounding in 2023.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Active malignancy (theoretical).

CJC-1295 (without DAC) | GH Secretagogue

FORMULA	C152H252N44O42S (modified 29-aa GHRH analog)
MOL. WEIGHT	3,367 Da
SEQUENCE	Modified GHRH(1-29) with Ala8, Gln15, Ala18, Gln22 substitutions
MECHANISM	Modified GHRH analog with increased resistance to enzymatic degradation vs. native GHRH. Stimulates pulsatile GH release. Without DAC linker, produces a natural-duration GH pulse (~30 min bioactive window), preserving physiological rhythm.
CLINICAL USES	Body composition, lean muscle preservation, fat loss, recovery, anti-aging. Most commonly stacked with Ipamorelin.
TYPICAL DOSE	100–300 mcg per injection, 1–3× daily
HALF-LIFE	~30 minutes (active window)
ROUTE	Subcutaneous injection
SIDE EFFECTS	Water retention, headache, transient flushing, carpal tunnel symptoms at high doses.
CLINICAL TRIALS	Published Phase I data (Teichman et al., 2006, JCEM) demonstrating dose-dependent GH and IGF-1 increases. No large-scale Phase III completed for body composition. FDA compounding ban November 2023.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Active neoplasms, acromegaly.

CJC-1295 (with DAC) | GH Secretagogue

FORMULA	C165H269N47O46S (GHRH analog with maleimido-propionyl-PEG DAC linker)
MOL. WEIGHT	~3,647 Da
SEQUENCE	Modified GHRH(1-29) + Drug Affinity Complex (DAC) maleimido linker covalently binding serum albumin
MECHANISM	The DAC linker allows covalent binding to circulating albumin, massively extending half-life from minutes to ~8 days. Produces sustained, non-pulsatile GH/IGF-1 elevation — physiologically different from natural pulsatile release.
CLINICAL USES	Long-acting GH elevation for body composition, anti-aging, metabolic support. Favored for once-weekly dosing convenience.
TYPICAL DOSE	1–2 mg subcutaneous, once weekly
HALF-LIFE	~6–8 days (albumin-bound)
ROUTE	Subcutaneous injection
SIDE EFFECTS	Water retention (more pronounced than no-DAC), arthralgias, carpal tunnel, potential blunting of natural GH rhythm with chronic use, insulin resistance.
CLINICAL TRIALS	Same Phase I/II data as no-DAC version (Teichman 2006). FDA compounding ban 2023 applies to both versions.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Diabetes, acromegaly, active cancer.

GHRP-6 | GH Secretagogue

FORMULA	C46H56N12O6 (His-D-Trp-Ala-Trp-D-Phe-Lys-NH2)
MOL. WEIGHT	873.0 Da
SEQUENCE	His-D-Trp-Ala-Trp-D-Phe-Lys-NH2
MECHANISM	Potent GHS-R1a agonist with strong ghrelin-mimicking activity. Produces robust GH release but simultaneously elevates ghrelin, causing marked appetite stimulation. Also elevates cortisol and prolactin more than Ipamorelin.
CLINICAL USES	GH optimization, muscle gain, appetite stimulation, recovery, cachexia treatment.
TYPICAL DOSE	100–300 mcg subcutaneous per injection
HALF-LIFE	~15–60 minutes
ROUTE	Subcutaneous injection
SIDE EFFECTS	Significant hunger spikes, water retention, cortisol elevation, prolactin elevation, potential gynecomastia with chronic use.
CLINICAL TRIALS	Multiple published pharmacological studies. Shown to increase GH 4–8-fold above baseline. Limited human RCT data for clinical endpoints.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Hypercortisolism, neoplasms.

GHRP-2 | GH Secretagogue

FORMULA	C45H55N9O6 (D-Ala-D-2-Nal-Ala-Trp-D-Phe-Lys-NH2)
MOL. WEIGHT	817.9 Da
SEQUENCE	D-Ala-D-2-Nal-Ala-Trp-D-Phe-Lys-NH2
MECHANISM	Potent GHS-R1a agonist. Similar mechanism to GHRP-6 but with somewhat reduced appetite stimulation. Still elevates cortisol and prolactin significantly with chronic use.
CLINICAL USES	GH optimization, body composition, recovery. Second-line to Ipamorelin due to side effect profile.
TYPICAL DOSE	100–300 mcg subcutaneous
HALF-LIFE	~1–2 hours
ROUTE	Subcutaneous injection
SIDE EFFECTS	Water retention, cortisol/prolactin elevation, hunger (less than GHRP-6), headache.
CLINICAL TRIALS	Published human PK/PD studies. FDA compounding ban 2023.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Hypercortisolism, active neoplasms.

MK-677 (Ibutamoren) | GH Secretagogue

FORMULA	C27H36N4O5S (non-peptide GHS / peptidomimetic small molecule)
MOL. WEIGHT	528.7 Da
SEQUENCE	Not a peptide — spiroindoline-derived small molecule; oral GHS
MECHANISM	Orally active ghrelin mimetic that binds GHS-R1a and GHS-R1b. Produces sustained, non-pulsatile elevation of GH and IGF-1. Unique for oral bioavailability — most GHS require injection. Also crosses blood-brain barrier influencing sleep architecture.
CLINICAL USES	GH/IGF-1 elevation, lean mass, sleep quality improvement (increases REM/slow-wave sleep), bone density, cognitive function, pediatric short stature (studied).
TYPICAL DOSE	10–25 mg orally, once daily
HALF-LIFE	~24 hours (stable GH elevation)
ROUTE	Oral
SIDE EFFECTS	Significant water retention, increased fasting blood glucose, increased appetite, joint pain, transient lethargy, potential long-term insulin resistance.
CLINICAL TRIALS	Multiple human RCTs. Nass et al. (2008, Ann Intern Med): sustained IGF-1 increases + improved body composition + enhanced sleep in elderly over 2 years. Murphy et al. (1998): improved cognition in elderly. Phase II for frailty/muscle wasting.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Type 2 diabetes, significant fluid retention, heart failure.

Tesamorelin | GH Secretagogue

FDA APPROVED

FORMULA	C221H366N72O67S (stabilized 44-aa GHRH analog with trans-3-hexenoic acid)
MOL. WEIGHT	5,135 Da
SEQUENCE	Trans-3-hexenoic acid-GHRH(1-44)-NH2 with chemical stabilization
MECHANISM	Stabilized GHRH analog with enhanced metabolic stability over native GHRH. Selectively reduces visceral adipose tissue (VAT) via GH/IGF-1 axis activation. Also shown to reduce amyloid burden and improve cognition.
CLINICAL USES	HIV-associated lipodystrophy (FDA-approved), visceral fat reduction, cognitive improvement in mild cognitive impairment (MCI), metabolic syndrome.
TYPICAL DOSE	2 mg subcutaneous daily
HALF-LIFE	~26 minutes
ROUTE	Subcutaneous injection
SIDE EFFECTS	Edema, arthralgias, myalgias, glucose intolerance, nausea, injection site reactions, carpal tunnel.
CLINICAL TRIALS	FDA Approved (2010, Egrifta). Falutz et al. (2010, NEJM): 15–20% VAT reduction in HIV patients. Baker et al. (2021, JAMA): improved executive function and reduced cortical amyloid in MCI — landmark cognitive trial.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	Pregnancy, lactation, active neoplasms, hypersensitivity, pituitary tumor.

4.2 Healing / Recovery

Peptides in the Healing / Recovery category

BPC-157 | Healing/Recovery

FORMULA	C62H98N16O22 (pentadecapeptide, 15 amino acids)
MOL. WEIGHT	1,419.5 Da
SEQUENCE	Gly-Glu-Pro-Pro-Pro-Gly-Lys-Pro-Ala-Asp-Asp-Ala-Gly-Leu-Val (partial sequence of BPC gastric protein)
MECHANISM	Activates VEGFR2, EGFR, and FAK signaling. Upregulates angiogenesis via VEGF pathway, promotes nitric oxide synthesis, modulates dopamine and serotonin systems, accelerates fibroblast migration and collagen deposition. Shows cytoprotection of gastric epithelium.
CLINICAL USES	Tendon and ligament repair, IBD/leaky gut/NSAID-induced GI damage, joint inflammation, wound healing, neuroprotection, PTSD, addiction recovery, liver/organ protection.
TYPICAL DOSE	250–500 mcg SC (systemic) or oral (GI); local injection near injury; 1–2× daily
HALF-LIFE	~4 hours
ROUTE	Subcutaneous, oral (GI conditions), or local injection
SIDE EFFECTS	Generally low toxicity. Mild nausea, dizziness. Theoretical: angiogenesis promotion in oncology patients (unproven in humans).
CLINICAL TRIALS	No published human RCTs — critical limitation. Extensive rat/animal studies. Phase II trials reported by Diagen/PL-10 (tendon, IBD) — unpublished. One of the most widely used research peptides with largest preclinical evidence base.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Active malignancy (theoretical precaution due to angiogenesis).

TB-500 (Thymosin Beta-4 fragment) | Healing/Recovery

FORMULA	C212H350N56O78S (synthetic fragment of full TB4)
MOL. WEIGHT	4,963 Da
SEQUENCE	Ac-Ser-Asp-Lys-Pro-Asp-Met-Ala-Glu-Ile-Glu-Lys-Phe-Asp-Lys-Ser-Lys-Leu-Lys-Lys-Thr-Glu-Thr-Gln-Glu-Lys-Asn-Pro-Leu-Pro-S...
MECHANISM	Sequesters G-actin to promote cell migration, angiogenesis, and tissue regeneration. The LKKTET motif activates integrin-linked kinase (ILK) for anti-apoptosis and cell survival signaling. Reduces inflammatory cytokines (IL-1β, TNF-α).
CLINICAL USES	Soft tissue repair (muscle, tendon, ligament), cardiac repair post-MI, hair regrowth promotion, neurological recovery, wound healing.
TYPICAL DOSE	2–2.5 mg SC twice weekly (loading); 1–2 mg biweekly (maintenance)
HALF-LIFE	~4–6 hours
ROUTE	Subcutaneous injection
SIDE EFFECTS	Minimal: fatigue, headache, mild nausea. Theoretical tumor angiogenesis concern.
CLINICAL TRIALS	Full Thymosin Beta-4 molecule: FDA Orphan Drug designation for epidermolysis bullosa; Phase II cardiac repair post-MI data (Goldstein et al.). TB-500 fragment: less direct human data, relies on parent molecule research.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Active cancer (theoretical precaution).

GHK-Cu (Copper Peptide) | Healing/Recovery

FORMULA	C14H23N6O4Cu (copper chelated tripeptide Gly-His-Lys)
MOL. WEIGHT	340.4 Da (peptide), ~403.9 Da with copper
SEQUENCE	Gly-His-Lys (tripeptide) + Cu^{2}■ chelation at histidine imidazole and lysine amino groups
MECHANISM	Naturally occurring in human plasma; declines significantly with age. Activates >4,000 human genes involved in: collagen/elastin synthesis, antioxidant defense (SOD, catalase), anti-inflammatory signaling (reduces TNF-α), DNA repair, and stem cell activation.
CLINICAL USES	Skin regeneration, wound healing, hair follicle stimulation, COPD/lung repair, nerve regeneration, bone repair, anti-aging (topical and systemic).
TYPICAL DOSE	1–10 mg SC weekly (systemic); topical: 0.1–1% formulations applied 1–2× daily
HALF-LIFE	~1–2 hours systemic
ROUTE	Subcutaneous injection or topical
SIDE EFFECTS	Topically: rare contact sensitization. Systemically: copper overload risk at very high doses. Otherwise excellent safety profile.
CLINICAL TRIALS	Pickart & Margolina (2018, Int J Mol Sci) — landmark review of genomic effects. Multiple human topical trials showing collagen synthesis improvement. COPD lung studies in animal models. Cosmeceutical studies showing measurable wrinkle reduction.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Wilson's disease (copper metabolism disorder).

4.3 Fat Loss / GLP-1

Peptides in the Fat Loss / GLP-1 category

Semaglutide | Fat Loss / GLP-1

FDA APPROVED

FORMULA	C187H291N45O59 (GLP-1 analog with C18 fatty diacid side chain via linker)
MOL. WEIGHT	4,113.6 Da
SEQUENCE	34-amino acid GLP-1 analog: His-Aib-Glu-Gly-Thr-Phe-Thr-Ser-Asp-Val-Ser-Ser-Tyr-Leu-Glu-Gly-Gln-Ala-Ala-Lys(C18-diacid-l...
MECHANISM	Potent GLP-1 receptor agonist. Increases glucose-dependent insulin secretion, suppresses glucagon, slows gastric emptying, and centrally reduces appetite via hypothalamic GLP-1R signaling. C18 fatty acid allows once-weekly dosing via albumin binding.
CLINICAL USES	Type 2 diabetes (Ozempic), obesity/overweight (Wegovy), cardiovascular risk reduction (MACE), NASH/MASLD (studied), Alzheimer's (studied).
TYPICAL DOSE	0.5–2.4 mg SC weekly (Wegovy); 2–14 mg oral daily (Rybelsus)
HALF-LIFE	~1 week (albumin binding)
ROUTE	Subcutaneous weekly / oral daily
SIDE EFFECTS	Nausea (30–50%), vomiting, diarrhea, constipation, GERD. Serious: pancreatitis, cholelithiasis, thyroid C-cell tumors (black box — animal data), diabetic retinopathy worsening, muscle loss concurrent with fat loss.
CLINICAL TRIALS	STEP-1 (Wilding et al., 2021, NEJM): 14.9% weight loss vs. 2.4% placebo. SUSTAIN-6: CV risk reduction. SELECT (2023): 20% CV risk reduction in non-diabetics with obesity. Multiple STEP trials across populations.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	MEN-2, personal/family history of medullary thyroid carcinoma, pregnancy.

Tirzepatide | Fat Loss / GLP-1

FDA APPROVED

FORMULA	C225H348N48O68 (dual GIP/GLP-1 agonist; 39-aa peptide with C20 fatty diacid)
MOL. WEIGHT	4,813.5 Da
SEQUENCE	39-amino acid peptide with Aib8 substitution; C20 fatty diacid via gamma-glutamic acid linker at Lys26; activates both G...
MECHANISM	First dual GIP (glucose-dependent insulinotropic polypeptide) and GLP-1 receptor agonist. GIP receptor activation in adipose tissue enhances lipid utilization; combined with GLP-1R central appetite suppression produces superior weight loss to monotherapy. Also improves insulin sensitivity via GIP action in bone and fat.
CLINICAL USES	Type 2 diabetes (Mounjaro), obesity (Zepbound), heart failure with obesity (studied), NASH (Phase III), sleep apnea.
TYPICAL DOSE	5–15 mg SC weekly
HALF-LIFE	~5 days
ROUTE	Subcutaneous weekly
SIDE EFFECTS	Similar to semaglutide: nausea, vomiting, diarrhea. Slightly higher GI tolerability. Same thyroid tumor black box warning. Muscle mass loss.
CLINICAL TRIALS	SURMOUNT-1 (Jastreboff et al., 2022, NEJM): 20.9% avg weight loss at 15mg. SURPASS trials for T2D. SURMOUNT-CVOT (2023): CV outcomes. FDA approved for T2D (2022) and obesity (2023). Largest weight loss in any pharmacotherapy trial.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	Same as semaglutide.

Liraglutide | Fat Loss / GLP-1

FDA APPROVED

FORMULA	C172H265N43O51 (GLP-1 analog with C16 fatty acid side chain)
MOL. WEIGHT	3,751.2 Da
SEQUENCE	34-aa GLP-1(7-37) analog with Arg34Lys substitution and palmitoyl-gGlu linker at Lys26
MECHANISM	Daily GLP-1R agonist. Albumin binding via C16 fatty acid extends half-life to ~13h. Same GLP-1R mechanism as semaglutide but requires daily dosing and produces somewhat less weight loss.
CLINICAL USES	Type 2 diabetes (Victoza), obesity (Saxenda), NASH (studied), cardiovascular risk reduction.
TYPICAL DOSE	1.2–1.8 mg SC daily (Victoza); 3 mg SC daily (Saxenda)
HALF-LIFE	~13 hours
ROUTE	Subcutaneous daily
SIDE EFFECTS	Nausea, vomiting, diarrhea, constipation. Same class black box thyroid warning.
CLINICAL TRIALS	LEADER trial: 13% CV risk reduction. SCALE trials: 5–8% weight loss. FDA approved for T2D (2010) and obesity (2014). First GLP-1 approved for obesity.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	MEN-2, medullary thyroid carcinoma history, pregnancy.

AOD-9604 | Fat Loss / GLP-1

FORMULA	C78H123N23O23S2 (HGH fragment 176-191, 16-aa)
MOL. WEIGHT	1,817.1 Da
SEQUENCE	Tyr-Leu-Arg-Ile-Val-Gln-Cys-Arg-Ser-Val-Glu-Gly-Ser-Cys-Gly-Phe (HGH residues 176-191)
MECHANISM	C-terminal fragment of human growth hormone. Retains the lipolytic activity of full HGH (via beta-3 adrenergic receptor activation) without growth-promoting effects or insulin desensitization. Does not bind GH receptors or elevate IGF-1.
CLINICAL USES	Targeted fat reduction (particularly visceral and abdominal), body composition, potential cartilage and bone repair.
TYPICAL DOSE	300–500 mcg SC daily, fasted state
HALF-LIFE	~2–3 hours
ROUTE	Subcutaneous injection
SIDE EFFECTS	Generally very well tolerated. Mild headache, transient fluid retention. No major adverse events in clinical trials.
CLINICAL TRIALS	Phase IIb/III by Metabolic Pharmaceuticals (Australia) — did not meet primary obesity endpoints vs. placebo in large trials despite promising Phase II data. FDA GRAS status granted for food use. Continues as research compound for body composition.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Pregnancy (caution).

4.4 Cognitive / Neuro

Peptides in the Cognitive / Neuro category

Semax | Cognitive/Neuro

FORMULA	C37H51N9O10S (Ac-Met-Glu-His-Phe-Pro-Gly-Pro-OH; 7-aa)
MOL. WEIGHT	813.9 Da
SEQUENCE	Ac-Met-Glu-His-Phe-Pro-Gly-Pro-OH (synthetic analog of ACTH 4-10 fragment + Pro-Gly-Pro extension)
MECHANISM	ACTH(4-7)PGP analog. Upregulates BDNF and NGF in hippocampus and cortex. Modulates dopamine, serotonin, and enkephalin systems. Activates Melanocortin receptor 4 (MC4R). Protects neurons from oxidative stress and ischemic damage.
CLINICAL USES	Cognitive enhancement, stroke rehabilitation, TBI recovery, ADHD (studied), optic nerve atrophy, anxiety, depression, PTSD, neuroprotection.
TYPICAL DOSE	100–900 mcg intranasal, 1–2× daily
HALF-LIFE	Active within ~15 min; neurological effects persist hours
ROUTE	Intranasal (primary); subcutaneous also used
SIDE EFFECTS	Generally well tolerated. Mild anxiety at high doses, nasal irritation, occasional irritability, potential tolerance with prolonged use.
CLINICAL TRIALS	Approved in Russia (MOH approved 1982, updated) for stroke, TBI, cognitive disorders. Multiple published Russian RCTs. Kupriyanova et al.: significant cognitive improvement in ADHD children. Eye disease studies showing optic nerve protection.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Active psychosis, epilepsy (caution).

Selank | Cognitive/Neuro

FORMULA	C33H57N11O9 (Thr-Lys-Pro-Arg-Pro-Gly-Pro; 7-aa tuftsin analog)
MOL. WEIGHT	751.9 Da
SEQUENCE	Thr-Lys-Pro-Arg-Pro-Gly-Pro (tuftsin Thr-Lys-Pro-Arg + Pro-Gly-Pro stabilizing extension)
MECHANISM	Synthetic tuftsin analog with anxiolytic and nootropic properties. Modulates GABAergic (via benzodiazepine site potentiation), serotonergic, and dopaminergic systems. Increases BDNF expression. Does not cause sedation, dependency, or withdrawal unlike benzodiazepines.
CLINICAL USES	Generalized anxiety disorder, depression, cognitive enhancement, stress, immune modulation, memory improvement.
TYPICAL DOSE	250–500 mcg intranasal, 1–3× daily
HALF-LIFE	Minutes (intranasal); extended effects via BDNF upregulation
ROUTE	Intranasal
SIDE EFFECTS	Exceptional safety profile. Mild fatigue, nasal irritation. No dependency, no withdrawal, no cognitive impairment.
CLINICAL TRIALS	Russian MOH approved for anxiety disorders. Comparative studies vs. benzodiazepines (Seredenin et al.): equivalent anxiolytic efficacy without sedation, dependency, or cognitive impairment. Immune studies showing T-cell modulation.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	None clearly established.

Dihexa | Cognitive/Neuro

FORMULA	C18H28N2O5 (angiotensin IV analog; N-hexanoic acid-Tyr-Ile-Pro-OH)
MOL. WEIGHT	340.4 Da
SEQUENCE	N-hexanoic acid-Tyr-Ile-Pro-OH (modified tripeptide; Angiotensin IV analog)
MECHANISM	Binds hepatocyte growth factor (HGF) and potentiates its receptor c-Met signaling. Promotes synaptogenesis (new synapse formation) in hippocampus and cortex. Estimated to be ~1 million times more potent than BDNF in cognitive enhancement assays.
CLINICAL USES	Cognitive enhancement, Alzheimer's disease (research), dementia, TBI recovery, PTSD (studied).
TYPICAL DOSE	10–30 mg oral or topical daily
HALF-LIFE	Unknown (estimated days)
ROUTE	Oral or transdermal
SIDE EFFECTS	Largely unknown in humans. Theoretical concerns about unregulated synaptogenesis. No significant adverse events in animal studies.
CLINICAL TRIALS	McCoy et al. (2013, J Pharmacol Exp Ther): massive spatial memory improvement in cognitively-impaired rats at sub-nanomolar doses. No published human clinical trials. University of Washington research origin.
EVIDENCE LEVEL	★★■■■
CONTRAINDICATIONS	Unknown; insufficient human data.

Cerebrolysin | Cognitive/Neuro

FORMULA	Complex mixture of low-molecular-weight neuropeptides from porcine brain
MOL. WEIGHT	~1,000 Da average (mixture of peptides <10 kDa)
SEQUENCE	Hydrolysate of purified porcine brain protein; contains BDNF-like, NGF-like, and CNTF-like peptide fractions
MECHANISM	Pleiotropic neuropeptide mixture. Acts like an endogenous neurotrophic cocktail: promotes neuronal survival (anti-apoptotic), neuroplasticity, neurogenesis, reduces amyloid toxicity, enhances cerebral glucose metabolism.
CLINICAL USES	Alzheimer's disease, vascular dementia, stroke rehabilitation, TBI, Parkinson's disease, cerebral palsy.
TYPICAL DOSE	5–30 mL IV or IM per day in cycles of 10–30 days
HALF-LIFE	Variable (complex mixture)
ROUTE	Intravenous (clinical doses) or intramuscular
SIDE EFFECTS	Seizures (rare), hypersensitivity reactions, injection site reactions, agitation, insomnia.
CLINICAL TRIALS	Over 100 clinical trials. Cochrane reviews: modest benefit in Alzheimer's and stroke. Approved in 34 countries (Germany, Austria, Russia, China, many others). Not FDA approved in US. Multiple ADAS-cog improvements in trials.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Epilepsy, porcine protein hypersensitivity, renal failure.

4.5 Sexual Health

Peptides in the Sexual Health category

PT-141 (Bremelanotide) | Sexual Health

FDA APPROVED

FORMULA	C50H68N14O10 (cyclic heptapeptide melanocortin analog)
MOL. WEIGHT	1,025.2 Da
SEQUENCE	Cyclic-(Nle4, Asp5, D-Phe7, Lys10) alpha-MSH(4-10): Ac-Nle-cyclo[Asp-His-D-Phe-Arg-Trp-Lys]-OH
MECHANISM	MC3R and MC4R agonist acting centrally in the hypothalamus and limbic system to increase sexual desire independent of vascular mechanisms. Activates dopaminergic and oxytocinergic pathways. Unlike PDE5 inhibitors (Viagra), works on central desire, not peripheral blood flow.
CLINICAL USES	Hypoactive Sexual Desire Disorder (HSDD) in premenopausal women (FDA-approved as Vyleesi). Off-label: male erectile dysfunction (particularly when PDE5 inhibitors fail), libido enhancement in both sexes.
TYPICAL DOSE	1.75 mg SC ~45 minutes before sexual activity; max 1× per 24h, 8× per month
HALF-LIFE	~2–4 hours
ROUTE	Subcutaneous injection
SIDE EFFECTS	Nausea (40% — most common), flushing, transient blood pressure increase (5–10 mmHg), hyperpigmentation with chronic use.
CLINICAL TRIALS	RECONNECT Phase III (Clayton et al., 2019, NEJM): significant improvement in FSFI desire domain and related personal distress. FDA approved August 2019 (Vyleesi). Male ED studies: >80% erection rate even when sildenafil failed.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Cardiovascular disease, uncontrolled hypertension, pregnancy. Avoid with nitrates.

4.6 Immune

Peptides in the Immune category

Thymosin Alpha-1 (TA1) | Immune

FORMULA	C129H215N33O55S (28-amino acid peptide; N-terminal thymus peptide)
MOL. WEIGHT	3,108.3 Da
SEQUENCE	Ac-Ser-Asp-Ala-Ala-Val-Asp-Thr-Ser-Ser-Glu-Ile-Thr-Thr-Lys-Asp-Leu-Lys-Glu-Lys-Lys-Glu-Val-Val-Glu-Glu-Ala-Glu-Asn-OH
MECHANISM	N-terminal fragment of prothymosin-alpha. Activates TLR9 signaling on dendritic cells, promotes Th1 cytokine profile (IFN-gamma, IL-2), enhances NK cell cytotoxicity, restores regulatory T-cell function. Acts as an endogenous thymic hormone.
CLINICAL USES	Chronic hepatitis B/C (approved in 35+ countries), HIV, cancer immunotherapy adjuvant, sepsis (Chinese guidelines), COVID-19, aging immune system restoration.
TYPICAL DOSE	1.6 mg SC 2× weekly
HALF-LIFE	~2 hours
ROUTE	Subcutaneous injection
SIDE EFFECTS	Excellent safety profile. Mild injection site reactions, transient flu-like symptoms (rare). No serious adverse events in 40+ RCTs.
CLINICAL TRIALS	Over 40 RCTs. Li et al. (2004, Lancet meta-analysis): reduced mortality in sepsis. Multiple HBV/HCV trials showing viral load reduction + immune normalization. Approved as Zadaxin in 35+ countries. COVID-19 studies: reduced mortality in severe cases.
EVIDENCE LEVEL	★★★★■
CONTRAINDICATIONS	Active autoimmune disease flares (use caution).

KPV (alpha-MSH fragment) | Immune

FORMULA	C16H26N4O5 (Lys-Pro-Val; C-terminal tripeptide of alpha-MSH)
MOL. WEIGHT	342.4 Da
SEQUENCE	Lys-Pro-Val (C-terminal tripeptide from alpha-MSH, residues 11-13)
MECHANISM	Potent anti-inflammatory peptide. Inhibits NF-kB nuclear translocation, reduces pro-inflammatory cytokines (IL-8, TNF-alpha, IL-6) in intestinal epithelium. Can be internalized by intestinal cells via specific receptors to act intracellularly.
CLINICAL USES	Inflammatory bowel disease (Crohn's, ulcerative colitis), intestinal inflammation, wound healing, skin inflammation.
TYPICAL DOSE	200–500 mcg oral or SC daily
HALF-LIFE	~2–4 hours
ROUTE	Oral (gut-targeted) or subcutaneous
SIDE EFFECTS	Very low toxicity profile in studies. Minimal side effects reported.
CLINICAL TRIALS	Preclinical studies showing potent IBD reduction in mouse models. Limited human data; ongoing research. No published Phase II/III human trials.
EVIDENCE LEVEL	★★■■■
CONTRAINDICATIONS	Unknown (insufficient data).

4.7 Anti-Aging

Peptides in the Anti-Aging category

Epitalon (Epithalon) | Anti-Aging

FORMULA	C14H22N4O9 (Ala-Glu-Asp-Gly; tetrapeptide)
MOL. WEIGHT	390.3 Da
SEQUENCE	Ala-Glu-Asp-Gly (tetrapeptide; synthetic analog of epithalamin from pineal gland)
MECHANISM	Activates telomerase enzyme (hTERT), extending telomere length in somatic cells. Regulates melatonin synthesis via pineal gland. Acts as epigenetic regulator — normalizes chromatin structure and gene expression patterns that shift with aging.
CLINICAL USES	Telomere extension, anti-aging, sleep regulation (via melatonin), cancer prevention (studied), longevity, immune senescence reversal.
TYPICAL DOSE	5–20 mg SC daily for 10–20 day cycles, 1–2× yearly
HALF-LIFE	~1–2 hours
ROUTE	Subcutaneous injection
SIDE EFFECTS	Very low toxicity in all studies. Mild injection site reactions. No serious adverse events reported. Limited long-term human data.
CLINICAL TRIALS	Extensive Khavinson research (St. Petersburg Institute of Bioregulation, Russia): telomerase activation in human cells (2003, Neuro Endocrinol Lett). 15+ year prospective study in elderly showing reduced mortality. Small human trials showing telomere length increases.
EVIDENCE LEVEL	★★■■■
CONTRAINDICATIONS	Active malignancy (theoretical — telomerase also active in cancer cells).

MOTS-c | Anti-Aging

FORMULA	C56H100N22O16S (16-aa mitochondrial-derived peptide)
MOL. WEIGHT	1,368.6 Da
SEQUENCE	Met-Arg-Trp-Gln-Glu-Met-Gly-Tyr-Ile-Phe-Tyr-Pro-Arg-Lys-Leu-Arg (encoded in 12S rRNA of mitochondrial DNA)
MECHANISM	Mitochondria-derived peptide encoded in mtDNA. Activates AMPK pathway (same mechanism as metformin). Regulates the folate cycle and purine synthesis. Translocates to nucleus under stress to activate stress-response genes. Declines with age.
CLINICAL USES	Metabolic health (insulin resistance, obesity), exercise mimetic (improves endurance without exercise), anti-aging, longevity, hot flash reduction (postmenopausal women — studied).
TYPICAL DOSE	5–10 mg SC weekly (research protocols)
HALF-LIFE	~days (not precisely characterized)
ROUTE	Subcutaneous injection
SIDE EFFECTS	Research compound; limited human safety data. No significant adverse events in studies to date.
CLINICAL TRIALS	Lee et al. (2015, Cell Metabolism): MOTS-c reverses diet-induced insulin resistance in mice via AMPK. Reynolds et al. (2021): improved physical performance in older mice. Phase I human trials initiated. Circulating MOTS-c levels correlate with longevity in centenarians.
EVIDENCE LEVEL	★★■■■
CONTRAINDICATIONS	Unknown (insufficient data).

Humanin | Anti-Aging

FORMULA	C72H119N21O19S (21-aa mitochondrial peptide)
MOL. WEIGHT	1,618 Da
SEQUENCE	Met-Ala-Pro-Arg-Gly-Phe-Ser-Cys-Leu-Leu-Leu-Leu-Thr-Ser-Glu-Ile-Asp-Leu-Pro-Val-Lys (encoded in 16S rRNA of mtDNA)
MECHANISM	Mitochondria-derived peptide acting via CNTFR/WSX-1/gp130 receptor complex. Inhibits neuronal apoptosis induced by amyloid-beta and other neurotoxins. Activates STAT3 and MAPK signaling for cell survival. Circulates in blood and declines with age.
CLINICAL USES	Neuroprotection, Alzheimer's prevention, anti-aging, cardiovascular protection, metabolic syndrome, male fertility (studied).
TYPICAL DOSE	Research stage; 2–5 mg SC in protocols
HALF-LIFE	~2–3 days
ROUTE	Subcutaneous injection
SIDE EFFECTS	Minimal in animal studies. No human safety profile established.
CLINICAL TRIALS	Human data: Muzumdar et al. (2009): Humanin levels lower in AD patients. Kim et al. (2018): lower levels in metabolic syndrome. Association studies show higher Humanin in centenarians. Phase I initiated.
EVIDENCE LEVEL	★★■■■
CONTRAINDICATIONS	Unknown.

4.8 Cardiovascular

Peptides in the Cardiovascular category

SS-31 (Elamipretide) | Cardiovascular

FORMULA	C32H52N8O6 (D-Arg-2'6'-dimethylTyr-Lys-Phe-NH2)
MOL. WEIGHT	640.8 Da
SEQUENCE	D-Arg-2',6'-dimethylTyr-Lys-Phe-NH2 (Szeto-Schiller tetrapeptide 31)
MECHANISM	Selectively partitions to inner mitochondrial membrane and binds cardiolipin — a unique phospholipid essential for electron transport chain (ETC) function. Stabilizes cardiolipin against peroxidation, restores ETC efficiency, and reduces mitochondrial ROS production.
CLINICAL USES	Heart failure with preserved ejection fraction (HFpEF), cardiac ischemia-reperfusion injury, renal ischemic injury, aging-related mitochondrial dysfunction.
TYPICAL DOSE	Clinical trial doses (0.05–0.25 mg/kg SC/IV)
HALF-LIFE	~2–3 hours
ROUTE	Subcutaneous or intravenous
SIDE EFFECTS	Well tolerated in trials. Injection site reactions, mild fatigue.
CLINICAL TRIALS	PROMISE-HF Phase IIa (Kadlec 2022): improved exercise tolerance and cardiac remodeling biomarkers in HFpEF. PROGRESS-HF Phase II (Penn et al., 2023, JACC HF): improved 6MWD, reduced NT-proBNP. FDA Fast Track designation. Multiple Phase II trials ongoing.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Investigational; consult trial protocols.

Nesiritide (BNP Recombinant) | Cardiovascular

FDA APPROVED

FORMULA	Recombinant human BNP; C134H221N39O50S (32-aa ring peptide via Cys10-Cys26 disulfide)
MOL. WEIGHT	3,464.0 Da
SEQUENCE	Ser-Pro-Lys-Met-Val-Gln-Gly-Ser-Gly-Cys-Phe-Gly-Arg-Lys-Met-Asp-Arg-Ile-Ser-Ser-Ser-Gly-Leu-Gly-Cys-Lys-Val-Leu-Arg-Arg-...
MECHANISM	Activates natriuretic peptide receptor-A (NPR-A) on vascular smooth muscle and endothelium, increasing cGMP and causing vasodilation. Reduces preload and afterload, increases sodium excretion, and suppresses the renin-angiotensin-aldosterone system.
CLINICAL USES	Acutely decompensated heart failure (hospital setting only).
TYPICAL DOSE	2 mcg/kg IV bolus then 0.01 mcg/kg/min infusion
HALF-LIFE	~18 minutes
ROUTE	Intravenous (hospital only)
SIDE EFFECTS	Hypotension (major), worsening renal function (controversial).
CLINICAL TRIALS	ASCEND-HF (2011, NEJM): no mortality benefit over standard care but dyspnea improvement. FDA approved 2001 as Natrecor. Post-market restrictions due to renal safety concerns.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	Cardiogenic shock, systolic BP <90 mmHg, pericardial constriction.

4.9 Skin / Cosmetic

Peptides in the Skin / Cosmetic category

Melanotan II (MTII) | Skin/Cosmetic

FORMULA	C50H69N15O9 (cyclic heptapeptide alpha-MSH analog)
MOL. WEIGHT	1,024.2 Da
SEQUENCE	Ac-Nle4-cyclo[Asp5-His6-D-Phe7-Arg8-Trp9-Lys10]-OH (cyclic; melanocortin receptor agonist)
MECHANISM	Non-selective melanocortin agonist. Activates MC1R (melanin synthesis, tanning), MC3R and MC4R (libido, appetite suppression, energy regulation). Potent and non-selective compared to PT-141 or Melanotan I.
CLINICAL USES	Skin tanning/pigmentation, erectile dysfunction (off-label), libido enhancement, appetite suppression.
TYPICAL DOSE	500 mcg–1 mg SC; start very low (250 mcg) due to potency
HALF-LIFE	~30–60 min active; tanning effects last days
ROUTE	Subcutaneous injection
SIDE EFFECTS	Significant: nausea (very common), facial flushing, spontaneous erections, darkening of existing moles, blood pressure changes. Serious concern: potential dysplastic nevus changes — documented cases of mole malignancy reported.
CLINICAL TRIALS	No regulatory approval anywhere globally. Phase I/II trials in 1990s for sexual dysfunction (Heaton et al.) and tanning (Dorr et al.). Discontinued due to safety signals. Not approved in any country. High-risk research compound.
EVIDENCE LEVEL	★★■■■
CONTRAINDICATIONS	History of melanoma or dysplastic nevi, cardiovascular disease, pregnancy. HIGH CAUTION required.

Snap-8 (Acetyl Octapeptide-3) | Skin/Cosmetic

FORMULA	C41H70N14O12 (acetylated octapeptide; Ac-EEMQRRAD-NH2)
MOL. WEIGHT	967.1 Da
SEQUENCE	Acetyl-Glu-Glu-Met-Gln-Arg-Arg-Ala-Asp-NH2 (extension of Argireline/Snap-6 sequence)
MECHANISM	Inhibits the SNARE complex formation (the molecular machinery required for acetylcholine vesicle fusion at neuromuscular junctions). Reduces neuronal signaling to facial muscles, decreasing repetitive facial contractions — topical 'botulinum-like' mechanism without injections.
CLINICAL USES	Reduction of expression lines (crow's feet, forehead lines), anti-aging cosmetic treatment.
TYPICAL DOSE	Topical 2× daily at 5–10% concentration
HALF-LIFE	N/A (topical)
ROUTE	Topical application only
SIDE EFFECTS	Excellent safety. Rare mild contact sensitization in very sensitive individuals.
CLINICAL TRIALS	Dragomirescu et al.: 15–30% reduction in wrinkle depth vs. vehicle in split-face studies. Industry-sponsored trials only. Not independently replicated at RCT level.
EVIDENCE LEVEL	★★■■■
CONTRAINDICATIONS	None established; patch test recommended for sensitive skin.

4.10 Pain

Peptides in the Pain category

Ziconotide (Prialt) | Pain

FDA APPROVED

FORMULA	C102H172N36O32S7 (25-aa omega-conotoxin MVIIA; cone snail peptide)
MOL. WEIGHT	2,639.1 Da
SEQUENCE	Cys-Lys-Gly-Lys-Gly-Ala-Lys-Cys-Ser-Arg-Leu-Met-Tyr-Asp-Cys-Cys-Thr-Gly-Ser-Cys-Arg-Ser-Gly-Lys-Cys-NH2 (3 disulfide bon...
MECHANISM	Selectively and reversibly blocks N-type voltage-gated calcium channels (Cav2.2) in spinal cord dorsal horn neurons. Prevents calcium-mediated neurotransmitter release at pain synapses. Highly potent — nanomolar affinity. Does not bind opioid receptors.
CLINICAL USES	Severe chronic pain refractory to opioids (cancer pain, HIV pain, failed back surgery), as intrathecal therapy.
TYPICAL DOSE	Starting 0.1 mcg/hour intrathecal; titrate to 0.4–0.8 mcg/hr (max 19.2 mcg/day)
HALF-LIFE	~4.6 hours (CSF)
ROUTE	Intrathecal only (implanted pump or external catheter)
SIDE EFFECTS	Serious: psychiatric symptoms (hallucinations, confusion, psychosis), dizziness, nausea, memory impairment, meningitis (catheter-related), narrow therapeutic window.
CLINICAL TRIALS	Phase III (Staats et al., 2004, JAMA): significant VAS pain reduction vs. placebo in cancer/AIDS pain. FDA approved December 2004 as Prialt. Multiple confirmatory studies. Specialist-only use.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	Psychiatric disorders, IV administration (fatal), history of psychosis, anticoagulant therapy.

4.11 Metabolic

Peptides in the Metabolic category

Octreotide | Metabolic

FDA APPROVED

FORMULA	C49H66N10O10S2 (8-aa cyclic somatostatin analog)
MOL. WEIGHT	1,019.3 Da
SEQUENCE	D-Phe-Cys-Phe-D-Trp-Lys-Thr-Cys-Thr-ol (disulfide bridge Cys2-Cys7; Sandoz)
MECHANISM	Potent somatostatin receptor (SSTR1-5) agonist. Inhibits release of GH, insulin, glucagon, gastrin, VIP, secretin, and other GI hormones. Reduces splanchnic blood flow. 45× more potent than native somatostatin with longer half-life.
CLINICAL USES	Acromegaly, carcinoid tumors (flushing/diarrhea), VIPomas, glucagonomas, variceal bleeding, diarrhea in AIDS/oncology, TSH-secreting tumors.
TYPICAL DOSE	50–200 mcg SC 3× daily (immediate-release); 10–30 mg IM monthly (Sandostatin LAR)
HALF-LIFE	~1.7 hours (immediate); 2 weeks (LAR depot)
ROUTE	Subcutaneous or IM (LAR)
SIDE EFFECTS	GI (nausea, diarrhea, abdominal pain), cholelithiasis (gallstones — 15–20% long-term), bradycardia, glucose dysregulation, hypothyroidism.
CLINICAL TRIALS	Multiple pivotal trials across indications. FDA approved 1988 for carcinoid/VIPoma; 1998 for acromegaly. Landmark trials for carcinoid tumor disease control.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	Known hypersensitivity. Monitor diabetes, gallbladder.

Oxytocin | Metabolic

FDA APPROVED

FORMULA	C43H66N12O12S2 (9-aa nonapeptide; Cys-Tyr disulfide ring)
MOL. WEIGHT	1,007.2 Da
SEQUENCE	Cys-Tyr-Ile-Gln-Asn-Cys-Pro-Leu-Gly-NH2 (disulfide bridge Cys1-Cys6; identical in all mammals)
MECHANISM	Natural hypothalamic neuropeptide. Activates OXTR (oxytocin receptor) throughout brain, uterus, breast, and cardiovascular system. Activates dopamine reward circuits and reduces amygdala fear response. Promotes social bonding, trust, and stress buffering.
CLINICAL USES	Labor induction (FDA-approved, Pitocin), uterine atony. Research/off-label: autism spectrum disorder (social cognition), social anxiety, PTSD, addiction, prosocial behavior enhancement.
TYPICAL DOSE	IV for obstetric (0.5–10 mU/min infusion). Intranasal research: 24–40 IU
HALF-LIFE	~1–5 minutes IV; ~45 minutes intranasal
ROUTE	IV (obstetric); intranasal (research/off-label)
SIDE EFFECTS	IV: uterine hyperstimulation, fetal distress, hypotension, hyponatremia. Intranasal: generally well tolerated; potential anxiogenic effect in some individuals at high doses.
CLINICAL TRIALS	FDA approved for obstetric use. Guastella et al. (2010, Biol Psychiatry): improved social recognition in ASD. Domes et al. (2007): enhanced mind-reading ability in healthy subjects. Multiple ongoing Phase II/III trials for ASD, schizophrenia.
EVIDENCE LEVEL	★★★■■
CONTRAINDICATIONS	Fetal malpresentation, uterine surgery history, fetal distress (IV). Cardiovascular disease (high-dose intranasal).

4.12 Hormonal

Peptides in the Hormonal category

Leuprolide (Leuprorelin) | Hormonal

FDA APPROVED

FORMULA	C59H84N16O12 (nonapeptide GnRH agonist)
MOL. WEIGHT	1,209.4 Da
SEQUENCE	pyroGlu-His-Trp-Ser-Tyr-D-Leu-Leu-Arg-Pro-NHEt (GnRH with D-Leu6, des-Gly10, Pro9-NHEt modifications)
MECHANISM	Potent GnRH receptor agonist. Initially stimulates LH/FSH (flare effect), but continuous administration causes receptor downregulation and desensitization, resulting in profound suppression of sex steroids (castrate levels of testosterone/estradiol) within weeks.
CLINICAL USES	Prostate cancer (androgen deprivation therapy), endometriosis, uterine fibroids, precocious puberty, IVF protocols, gender-affirming care.
TYPICAL DOSE	1 mg SC daily; depot 3.75–22.5 mg IM monthly to 6-monthly
HALF-LIFE	~3 hours (SC); depot release 1–6 months
ROUTE	SC daily or IM depot
SIDE EFFECTS	Hot flashes, sexual dysfunction, bone density loss, fatigue, mood changes, cardiovascular risk (long-term in men), metabolic syndrome.
CLINICAL TRIALS	Pivotal prostate cancer trials (Crawford et al.): equivalent to surgical castration. Multiple endometriosis trials. FDA approved 1985. Generic available. NCCN guideline standard of care for prostate cancer.
EVIDENCE LEVEL	★★★★★
CONTRAINDICATIONS	Undiagnosed vaginal bleeding, pregnancy (causes fetal harm).

5 Clinical Trials Summary

Key Published Studies & Evidence Level

Peptide	Study	Year	Journal	N	Key Finding	Status
Semaglutide	STEP-1 (Wilding et al.)	2021	NEJM	1,961	14.9% weight loss vs 2.4% placebo over 68 weeks	★★★★★ — FDA Approved
Tirzepatide	SURMOUNT-1 (Jastreboff)	2022	NEJM	2,539	20.9% weight loss at 15mg over 72 weeks — largest ever	★★★★★ — FDA Approved
Semaglutide CV	SELECT (Lincoff)	2023	NEJM	17,604	20% MACE reduction in non-diabetic obese patients	★★★★★ — Label update
Liraglutide	LEADER Trial	2016	NEJM	9,340	13% CV risk reduction; 3% weight loss avg	★★★★★ — FDA Approved
Tesamorelina	PROM Trial (Falutz)	2010	NEJM	412	15–20% VAT reduction in HIV lipodystrophy patients	★★★★★ — FDA Approved
Tesamorelina	Baker et al. (MCI)	2021	JAMA	152	Improved executive function + reduced amyloid burden in MCI	★★★★■ — Label extension studied
Bremelanotide	RECONNECT (Clayton)	2019	NEJM	1,267	Significant FSFI desire improvement vs placebo (p<0.0001)	★★★★★ — FDA Approved
Ziconotide	Staats et al.	2004	JAMA	111	Significant VAS reduction in refractory cancer/AIDS pain	★★★★★ — FDA Approved
MK-677	Nass et al.	2008	Ann Intern Med	65	Sustained IGF-1 increase + body composition + sleep in elderly	★★★★■ — Investigational
CJC-1295	Teichman et al.	2006	JCEM	21	Dose-dependent GH/IGF-1 increases lasting 6+ days	★★★★■ — Investigational
Thymosin Alpha-1	Li et al. meta-analysis	2004	Lancet	361	Reduced mortality in sepsis (Chinese multicenter RCTs)	★★★★■ — Approved 35+ countries
SS-31	PROGRESS-HF (Penn)	2023	JACC HF	43	Improved exercise tolerance + cardiac remodeling in HFpEF	★★★■■ — Phase II
MOTS-c	Lee et al.	2015	Cell Metabolism	N/A	Reversed diet-induced insulin resistance via AMPK in mice	★★■■■ — Preclinical
Epitalon	Khavinson et al.	2003	Neuro Endocrinol Lett	—	Telomerase activation + telomere lengthening in human cells	★★■■■ — Russian research
Dihexa	McCoy et al.	2013	J Pharmacol Exp Ther	N/A	Massive spatial memory improvement in impaired rats	★★■■■ — Preclinical
Cerebrolysin	Gauthier et al. Cochrane	2013	Cochrane	1,549	Modest global clinical impression improvement in Alzheimer's	★★★★■ — Approved in Europe/Asia
Oxytocin (ASD)	Guastella et al.	2010	Biol Psychiatry	29	Improved social recognition in ASD adults vs placebo	★★★■■ — Phase II ongoing
Semax (stroke)	Kuznetsov et al.	2014	J Neurol Sci	82	Improved NIHSS scores at 30 days vs standard care in ischemic stroke	★★★★■ — Approved in Russia
BPC-157	Chang et al. review	2020	J Physiol Pharmacol	N/A	Review of tendon, GI, nerve repair in animal models — no human RCTs	★★★■■ — Preclinical dominant
Leuprolide	Crawford et al.	1989	NEJM	135	Equivalent to orchiectomy for prostate cancer androgen suppression	★★★★★ — FDA Approved 1985

Evidence Key: ★★★★★ = FDA Approved with multiple Phase III RCTs | ★★★★ = Human clinical data (Phase I–II) | ★★★ = Limited human or strong animal | ★★ = Preclinical / in vitro only

6 Safety & Risk Reference

Side Effects · Contraindications · Risk Rating

Peptide	Common Side Effects	Serious Risks	Contraindications	Risk Rating
Semaglutide / Tirzepatide	Nausea, vomiting (30–50%), diarrhea	Pancreatitis, thyroid C-cell tumors, cholelithiasis	MEN-2, medullary thyroid Ca, pregnancy	MODERATE — Manage with dose titration
Ipamorelin / Sermorelin	Injection site, water retention, headache	Pituitary desensitization (chronic overuse)	Active neoplasms, untreated hypothyroidism	LOW — Well tolerated
CJC-1295 (DAC)	Water retention, arthralgias, carpal tunnel	Insulin resistance, blunted GH rhythm	Diabetes, acromegaly, active cancer	LOW-MODERATE
GHRP-6 / GHRP-2	Hunger, water retention, flushing	Elevated cortisol & prolactin; potential gynecomastia	Hypercortisolism, neoplasms	MODERATE — Monitor hormones
MK-677	Water retention, high fasting glucose, hunger, joint pain	Insulin resistance; unknown long-term IGF-1 risks	T2D, heart failure, severe fluid retention	MODERATE
BPC-157	Mild nausea, dizziness	Theoretical tumor angiogenesis (unproven)	Active malignancy (caution)	VERY LOW
TB-500	Mild fatigue, headache	Theoretical tumor angiogenesis	Active malignancy (caution)	LOW
GHK-Cu	Topical: rare irritation	Copper overload at very high systemic doses	Wilson's disease	VERY LOW
PT-141	Nausea (40%), flushing, BP increase	Hypertensive crisis (rare), mole hyperpigmentation	Cardiovascular disease, uncontrolled HTN	MODERATE
Semax	Nasal irritation, mild anxiety	Unknown long-term neurological effects	Active psychosis, epilepsy	LOW
Selank	Mild fatigue, nasal irritation	None documented	None established	VERY LOW
Melanotan II	Nausea, flushing, spontaneous erections, mole darkening	Melanoma risk from nevus changes (documented cases)	Melanoma history, CV disease	HIGH — Use extreme caution
Ziconotide	Dizziness, confusion, nausea	Psychosis, hallucinations, meningitis, coma	Psychiatric disorders, IV use	HIGH — Specialist only
Thymosin Alpha-1	Mild injection site, transient flu	Rare hypersensitivity; autoimmune flare	Active severe autoimmune disease	VERY LOW
Epitalon	Minimal; injection site	Unknown long-term (theoretical cancer via telomerase)	Active malignancy	LOW
Cerebrolysin	Dizziness, agitation, insomnia	Seizures (rare), anaphylaxis	Epilepsy, porcine protein allergy	LOW-MODERATE
Octreotide	GI symptoms, gallstones (15–20% long-term)	Cholelithiasis, bradycardia, glucose dysregulation	Hypersensitivity	LOW-MODERATE — Monitor gallbladder
Leuprolide	Hot flashes, sexual dysfunction, bone loss	Cardiovascular risk, metabolic syndrome (long-term)	Undiagnosed vaginal bleeding, pregnancy	MODERATE — Long-term monitoring

7 Administration Routes Guide

How & Where Peptides Are Delivered

Route	Bioavailability	Key Peptides	Practical Notes
Subcutaneous (SC)	70–90%	Most peptides (BPC-157, Ipamorelin, TB-500, TA1, semaglutide, etc.)	29–31G insulin syringe; 0.5–1 inch needle; rotate sites (abdomen, thigh, upper arm); refrigerate reconstituted peptide; use bacteriostatic water
Intramuscular (IM)	~80–95%	TB-500, Cerebrolysin (large volumes), Leuprolide depot, Octreotide LAR	Larger volume tolerated vs SC; slightly more painful; 25G needle typical; depot formulations last 1–6 months
Intranasal	40–70%	Semax, Selank, Oxytocin, PT-141 (some protocols)	Nasal mucosa absorbs small lipophilic peptides; unique CNS access via olfactory nerve (bypasses BBB for brain-targeted peptides); convenient — no injection needed
Oral	Variable (<5% systemic; high GI locally)	BPC-157 (GI conditions), MK-677 (oral GHS), GLP-1 oral (Rybelsus — special formulation)	Most peptides degraded by stomach acid/proteases; oral BPC-157 effective locally for GI conditions without systemic effect; Rybelsus uses SNAC absorption enhancer
Topical / Transdermal	5–30%	GHK-Cu (cosmetic), Snap-8/Argireline, Dihexa, Melanotan (tanning applications)	Good for local dermal effects; penetration enhancers (DMSO, nanoparticles) improve delivery; useful for musculoskeletal injuries locally
Intravenous (IV)	100% (immediate)	Cerebrolysin (clinical doses), Nesiritide, Ziconotide loading, Octreotide drip	Clinical setting only; fastest onset; highest risk of adverse reactions; requires sterile preparation and medical supervision
Intrathecal	~100% to CSF	Ziconotide (Prialt) — only FDA-approved intrathecal peptide	Delivered directly into cerebrospinal fluid via implanted pump or external catheter; bypasses blood-brain barrier entirely; specialist-only procedure; extreme caution required

Storage & Reconstitution

Lyophilized (freeze-dried) peptide powder: store at 4°C (refrigerator); stable 12+ months. After reconstituting with bacteriostatic water: refrigerate immediately; use within 30 days. Never freeze reconstituted peptides (ice crystals damage the molecular structure). Bacteriostatic water (0.9% benzyl alcohol) is preferred over sterile water for multi-use vials. Typical dilution: 2–3 mL bacteriostatic water per 5 mg vial. Protect from light.

8 Legal & Regulatory Status

FDA · Compounding · Research · International

Peptide	FDA Status	Brand Name	US Availability	Key Notes
Semaglutide	■ FDA APPROVED	Ozempic / Wegovy / Rybelsus	Rx only	Compounding restricted post-shortage resolution 2024
Tirzepatide	■ FDA APPROVED	Mounjaro / Zepbound	Rx only	Fastest-approved GLP-1; compounding scrutinized
Liraglutide	■ FDA APPROVED	Victoza / Saxenda	Rx only	Biosimilars in development
Tesamorelin	■ FDA APPROVED (limited)	Egrifta / Egrifta SV	Rx — HIV indication	Off-label use widespread but Rx required
Bremelanotide	■ FDA APPROVED	Vyleesi	Rx — women only official	Men use off-label
Ziconotide	■ FDA APPROVED	Prialt	Rx — specialist only	Intrathecal only; hospital/pain clinic
Octreotide	■ FDA APPROVED	Sandostatin	Rx	Multiple indications; generic available
Leuprolide	■ FDA APPROVED	Lupron Depot	Rx	Generic available; standard of care
Oxytocin	■ FDA APPROVED (obstetric)	Pitocin	Rx (IV obstetric)	Intranasal off-label; IV approved
Nesiritide	■ FDA APPROVED (hospital)	Natrecor	Hospital Rx only	Post-ASCEND use restricted
Sermorelin	■■ Compounding only	Geref (discontinued)	Magistral pharmacy	Original approval withdrawn; legal via compounding
Ipamorelin	■ BANNED compounding	N/A	Research chemical only	FDA compounding ban November 2023
CJC-1295	■ BANNED compounding	N/A	Research chemical only	FDA compounding ban November 2023
GHRP-2 & GHRP-6	■ BANNED compounding	N/A	Research chemical only	FDA compounding ban November 2023
BPC-157	■ Not approved (gray area)	N/A	Research chemical	FDA considering restriction; currently gray area
TB-500	■ Not approved	N/A	Research chemical	Gray area; legal for research
MK-677	■ Not approved	N/A	Research chemical / supplement gray area	Sold as SARMs research chemical
Thymosin Alpha-1	■ Not FDA approved	Zadaxin (international)	Research chemical in US	Approved in 35+ countries; widely used in oncology internationally
Melanotan II	■ NOT APPROVED — Illegal to sell	N/A	Black market only	No country has approved; high-risk compound
Semax	■ Not FDA approved	N/A	Import gray area	Approved in Russia; legal to possess in US; import uncertain
Selank	■ Not FDA approved	N/A	Import gray area	Approved in Russia; similar status to Semax
Epitalon	■ Not approved	N/A	Research chemical	Research use only
GHK-Cu	■ Legal (topical cosmetic)	Multiple cosmetic brands	OTC topical / SC research	Topical fully legal; SC = research compound

AOD-9604	■ Not approved	N/A	Research chemical	GRAS for food; not approved as drug

2023 FDA Compounding Policy — Major Change

In November 2023, the FDA removed Ipamorelin, CJC-1295, GHRP-2, GHRP-6, Hexarelin, and several other peptides from the 503A/503B bulk drug substances list, effectively banning licensed compounding pharmacies from preparing them. This is the most significant regulatory change affecting the peptide industry in decades. Physicians who previously prescribed these compounds via compounding pharmacies must now use alternative approaches. The law continues to evolve — verify current status with a healthcare provider.

9 Top 10 Best-Reviewed Peptide Stores Online

Research Vendors — Ranked by Reviews, Purity & Transparency · 2025/2026

Important Notice — Research Chemicals Only

All vendors listed below sell peptides strictly labeled as research chemicals for laboratory use. They are NOT licensed pharmacies and their products are not intended for human consumption. Rankings are based on aggregated community reviews (Trustpilot, Reddit, Outliyr, Muscle & Brawn, ResearchPeptides.org), third-party testing standards, and purity documentation. Always verify current status — this market changes rapidly. For therapeutic use, use only FDA-approved products from licensed pharmacies with a valid prescription.

#1 Peptide Sciences

★★★★★ | Trustpilot: 4.8 / 5

WEBSITE	peptidesciences.com	**LOCATION**	USA
PURITY	≥99%	**TESTING**	HPLC + Mass Spectrometry (in-house + third-party). COAs available per batch.
CATALOG	70+ peptides; research-only focus; no SARMs — pure peptide specialization	**BEST FOR**	Professional researchers and labs prioritizing maximum documentation and verified purity
■ STRENGTHS	Most established US vendor. WHO/GMP & ISO 9001:2015 certified manufacturers. Batch-specific COAs available before purchase. SSL-encrypted ordering. Industry gold standard for purity documentation. Frequently cited by researchers as #1.	**■ LIMITATIONS**	Pricing 15–25% above industry average. No human-use framing. Returns not accepted.

#2 Core Peptides

★★★★★ | Trustpilot: 4.9 / 5

WEBSITE	corepeptides.com	**LOCATION**	USA
PURITY	≥99%	**TESTING**	Third-party HPLC + MS. SafeCert Labs (CLIA-certified) COAs. Endotoxin testing.
CATALOG	60+ peptides; custom synthesis available; cGMP facility; free shipping on orders $200+	**BEST FOR**	Researchers who want the highest purity combined with fast domestic shipping
■ STRENGTHS	Consistently top-ranked for purity. Same-day shipping on US orders. Praised for responsive customer service. cGMP-produced. Transparent batch documentation. One of the few with SafeCert CLIA-certified COAs publicly listed.	**■ LIMITATIONS**	Slightly smaller catalog than Pure Rawz. Some products sell out quickly.

#3 Limitless Biotech (formerly Limitless Life Nootropics)

★★★★■ | Trustpilot: 3.9 / 5 (148 reviews — mixed)

WEBSITE	limitlesslifenoootropics.com	**LOCATION**	Gulf Breeze, Florida, USA
PURITY	≥98%	**TESTING**	Third-party independent testing; published COAs. Endorsed by Dr. Bill Lawrence, Steve Horvath, Ben Greenfield, Jay Campbell.
CATALOG	34+ peptides including rare formats (nasal sprays, oral drops). GLP-1s, Semax, Selank, GHK-Cu.	**BEST FOR**	Biohackers and longevity enthusiasts wanting a wide variety including rare delivery formats
■ STRENGTHS	Ranked #1 overall by Outliyr (2025). US small business. Expert-endorsed. Rare peptide formats. Fast US shipping. Strong community following. Transparent owner accessible via social media. VIP membership for exclusive products.	**■ LIMITATIONS**	Mixed Trustpilot reviews — some customer service complaints. Some products VIP-only (requires membership). International shipping expensive.

#4 Pure Rawz

★★★★■ | Trustpilot: 4.5 / 5

WEBSITE	purerawz.co	**LOCATION**	USA
PURITY	≥95–99%	**TESTING**	First and third-party lab testing. COAs available (some gaps noted by reviewers).
CATALOG	69+ peptides — largest catalog in the market. Peptide blends (13+). Capsules, liquid, powder formats. SARMs, nootropics too.	**BEST FOR**	Researchers needing the widest possible catalog or hard-to-find peptide compounds
■ STRENGTHS	Massive catalog including GLP-1 class, myostatin, GDF-8, EPO and rare compounds. Multiple formats (capsule/liquid/powder). Strong community presence. Good for researchers needing hard-to-find peptides. Social media coupons available.	**■ LIMITATIONS**	Some COA gaps for certain products. Customer service response time varied. Past aggressive marketing drew criticism. Some complaints about response delays.

#5 Pure Peptide Labs

★★★★★ | Trustpilot: 5.0 / 5 (1,494 reviews — outstanding)

WEBSITE	purepeptidelabs.com	**LOCATION**	USA
PURITY	≥98%	**TESTING**	Batch-specific COAs provided. Third-party quality verification.
CATALOG	40+ peptides. Free bacteriostatic water included with orders. Donation-based business model.	**BEST FOR**	Customers who prioritize exceptional service + community trust over catalog breadth
■ STRENGTHS	Highest Trustpilot volume with near-perfect score (1,494 reviews). Exceptional customer service (multiple reviews praising staff by name). Free bac water included. Unique donation model. Fast delivery. Strong community loyalty — 'my go-to peptide company' cited repeatedly.	**■ LIMITATIONS**	Smaller catalog than Pure Rawz. Less established history than Peptide Sciences.

#6 Science.bio

★★★★■ | Trustpilot: 4.5 / 5

WEBSITE	science.bio	**LOCATION**	USA
PURITY	≥98–99%	**TESTING**	Premium third-party testing; detailed COAs. Works directly with pharma scientists.
CATALOG	Focused catalog; metabolic peptides, nootropics, longevity compounds. Spray formats.	**BEST FOR**	Labs and researchers requiring compliance-level documentation for institutional work
■ STRENGTHS	Premium documentation preferred by labs. Pharma-scientist collaboration. Strict compliance documentation ideal for institutional researchers. Relaunched with stronger quality focus post-2023 hiatus.	**■ LIMITATIONS**	Smaller peptide selection vs competitors. Limited catalog still recovering post-relaunch.

#7 Biotech Peptides

★★★★★ | Trustpilot: 5.0 / 5 (334 reviews)

WEBSITE	biotechpeptides.com	**LOCATION**	USA
PURITY	≥98%	**TESTING**	Third-party HPLC testing. COAs available per product.
CATALOG	50+ peptides; solid catalog of common research peptides; competitive pricing.	**BEST FOR**	Value-conscious researchers wanting proven quality at competitive prices
■ STRENGTHS	Perfect Trustpilot score with 334 reviews. Consistently praised for product quality. Competitive pricing relative to Peptide Sciences. Reliable shipping. Multiple positive reviews noting visible research effects.	**■ LIMITATIONS**	Less prominent in professional research circles than Peptide Sciences. Fewer rare or novel peptides vs. Pure Rawz or Limitless.

#8 Swiss Chems

★★★★■ | Trustpilot: 4.3 / 5

WEBSITE	swisschems.is	**LOCATION**	International (ships to USA)
PURITY	≥97–99%	**TESTING**	HPLC + MS transparent lab reports. COAs listed per product.
CATALOG	Extensive: peptides, SARMs, PCT, male enhancement. Diverse product types. Lyophilized powder.	**BEST FOR**	Researchers wanting a broad catalog including SARMs + peptides from a single source
■ STRENGTHS	Transparent lab reports. Diverse product catalog. Reasonable pricing. Widely cited in top-10 lists. International operation with US shipping. Accepts diverse payment methods including crypto.	**■ LIMITATIONS**	Quality noted as 'fallen' vs prior years by some reviewers (2024). International operation = longer shipping. Non-US manufacturing.

#9 Raw Amino

★★★★■ | Trustpilot: 4.4 / 5

WEBSITE	rawamino.com	**LOCATION**	USA
PURITY	≥99%	**TESTING**	Strict third-party testing. GMP manufacturing. Highly regarded for purity reliability.
CATALOG	40+ peptides; GMP-certified manufacturing; precision-focused catalog.	**BEST FOR**	Researchers who need maximum purity assurance with GMP-grade manufacturing
■ STRENGTHS	Highly trusted for strict testing and purity reliability. GMP-certified manufacturing. Praised in multiple 2025 best-vendor roundups. Excellent for researchers who prioritize purity above catalog breadth.	**■ LIMITATIONS**	Smaller catalog. Less community presence than Limitless or Pure Rawz.

#10 Paradigm Peptides

★★★★■ | Trustpilot: 4.2 / 5

WEBSITE	paradigmpeptides.com	**LOCATION**	USA
PURITY	≥98%	**TESTING**	Third-party testing via Janoshik lab. Recent batch COAs (May 2025). Lyophilized powder only.
CATALOG	50+ peptides; lyophilized powder format; competitive pricing; all major research peptides.	**BEST FOR**	Budget-conscious researchers who are comfortable reconstituting lyophilized peptides
■ STRENGTHS	Reputable US vendor with consistent third-party Janoshik testing. Competitive prices. Lyophilized format for maximum stability. Active Reddit presence. Good for budget-conscious researchers.	**■ LIMITATIONS**	Lyophilized only — requires reconstitution (no pre-mixed options). Smaller than top-tier vendors. Less documentation depth than Peptide Sciences.

Quick Comparison Summary

Rank	Store	Trustpilot	Purity	Best For
#1	Peptide Sciences	4.8 ★	≥99%	Max purity + documentation
#2	Core Peptides	4.9 ★	≥99%	Purity + fast US shipping
#3	Limitless Biotech	3.9 ★	≥98%	Wide variety + rare formats
#4	Pure Rawz	4.5 ★	≥95–99%	Largest catalog
#5	Pure Peptide Labs	5.0 ★	≥98%	Customer service + trust
#6	Science.bio	4.5 ★	≥99%	Lab/institutional compliance
#7	Biotech Peptides	5.0 ★	≥98%	Value + quality balance
#8	Swiss Chems	4.3 ★	≥97%	Peptides + SARMs combo
#9	Raw Amino	4.4 ★	≥99%	GMP manufacturing

#10	Paradigm Peptides	4.2 ★	≥98%	Budget-friendly lyophilized

How to Verify Any Peptide Vendor — Buyer's Checklist

Before purchasing from any vendor: (1) Request a batch-specific COA — not a generic one. It must show HPLC chromatogram + method + mass spectrometry with matching molecular weight. (2) Check Trustpilot and Reddit (r/Peptides, r/PeptideVendors) for recent reviews. (3) Start with a small order to test quality before bulk purchasing. (4) Avoid any vendor claiming products are 'for human use' — this is a legal red flag. (5) Check Finnrick Analytics (finnrick.com/vendors) — independent lab testing database with 176 vendor quality ratings based on blind sample testing.

10 Medical Disclaimer

Important Information Before Using This Guide

IMPORTANT MEDICAL AND LEGAL DISCLAIMER

This Comprehensive Peptide Guide is compiled solely for educational and informational purposes. The information contained herein does NOT constitute medical advice, diagnosis, or treatment recommendations for any individual. The field of therapeutic peptides includes FDA-approved medications, investigational compounds, research chemicals, and substances with unclear regulatory status — all of which have fundamentally different legal standing and safety profiles.

Consult a Healthcare Provider

Never initiate any peptide protocol without consulting a licensed physician, preferably one experienced in peptide therapy, endocrinology, or the relevant specialty. Self-administering any compound based on online research carries serious health risks.

Research Compounds Are Not Drugs

Many peptides listed here are classified as research chemicals — they have not undergone the rigorous safety and efficacy testing required for human drug approval. Unknown risks exist, including long-term effects, drug interactions, and contamination from unregulated suppliers.

Purity & Quality Concerns

Research peptides purchased online may be mislabeled, underdosed, overdosed, or contaminated with bacterial endotoxins or other harmful substances. There is no FDA oversight of these products. If peptides are used, obtain them only from licensed compounding pharmacies with USP verification.

Legal Compliance

The regulatory status of peptides varies by country and changes frequently. Users are responsible for understanding and complying with local laws. The 2023 FDA compounding ban dramatically changed the US landscape for GH-related peptides.

Not for Human Use Statements

Many compounds listed as 'research chemicals' carry explicit 'not for human use' labeling. These statements have legal meaning and reflect the absence of safety data necessary for human administration.

Evidence Quality

The evidence stars (★) throughout this guide reflect the author's assessment of published data quality. Even high-evidence FDA-approved peptides carry risks. Low-evidence compounds should be approached with corresponding caution.

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437943)

Triple Agonist: GLP-1R + GIP-R + GCGR · The Most Potent Weig

MOLECULAR FORMULA	C194H330N52O63 (GLP-1/GIP/glucagon triple receptor agonist peptide; 39 amino acids)
MOLECULAR WEIGHT	4,342 Da
SEQUENCE	Modified 39-aa peptide with fatty acid C20 diacid conjugation via linker at Lys20; activates GLP-1R, GIP-R, and GCGR simultaneously
MECHANISM OF ACTION	First-in-class TRIPLE receptor agonist. Acts simultaneously on: • GLP-1R: Suppresses appetite centrally, slows gastric emptying, increases insulin secretion • GIP-R: Enhances adipose tissue lipid utilization, improves insulin sensitivity • GCGR: Glucagon receptor activation increases energy expenditure and hepatic fat clearance The glucagon component is what separates Retatrutide from Tirzepatide — it adds a third distinct mechanism that increases basal metabolic rate, making it the most potent weight loss agent ever tested in clinical trials.
CLINICAL USES	Obesity / overweight (Phase 3 trials ongoing), Type 2 diabetes, MASLD/NASH (fatty liver disease), metabolic syndrome, cardiovascular risk reduction. Not yet FDA approved — Phase 3 TRIUMPH trials underway (2024-2026).
DOSING PROTOCOL	Phase 2 trial doses: 1mg, 4mg, 8mg, 12mg SC weekly Titration schedule: Start 2mg SC weekly × 4 weeks → 4mg × 4 weeks → 8mg × 4 weeks → 12mg (max dose) SLOWER titration recommended vs Semaglutide due to more pronounced GI side effects at higher doses
HALF-LIFE	~6 days (albumin-bound via C20 fatty acid — similar to Tirzepatide)
ROUTE	Subcutaneous injection, once weekly
EXPECTED RESULTS	Phase 2 SURMOUNT-like trial (Jastreboff et al., 2023, NEJM): • 1mg: 8.7% body weight loss at 24 weeks • 4mg: 17.3% body weight loss at 24 weeks • 8mg: 22.8% body weight loss at 24 weeks ← Best-in-class • 12mg: 24.2% body weight loss at 24 weeks ← Unprecedented in any drug trial For comparison: Semaglutide 2.4mg achieves ~15% at 68 weeks; Tirzepatide 15mg achieves ~21% at 72 weeks. Retatrutide achieved 24% at just 24 weeks — roughly double the speed of competitors.
SIDE EFFECTS	Common: Nausea (50-60%), vomiting, diarrhea, constipation — more pronounced than Tirzepatide Serious (same class warnings): Thyroid C-cell tumor risk (black box warning, rodent data), pancreatitis, gallbladder disease, tachycardia (from glucagon receptor component) Notable addition vs GLP-1 only: Elevated heart rate (glucagon effect on GCGR) — monitor cardiovascular
CLINICAL TRIALS	PHASE 2 KEY TRIAL: Jastreboff et al. (2023, NEJM) — 338 adults with obesity, 24 weeks Results: Up to 24.2% body weight loss — highest ever recorded in a pharmacological trial PHASE 3 TRIUMPH PROGRAM (ongoing 2024-2026): • TRIUMPH-1: Obesity without diabetes (2,000+ patients) • TRIUMPH-2: Obesity with Type 2 diabetes • TRIUMPH-3: Cardiovascular outcomes • TRIUMPH-4: MASLD/NASH Expected FDA submission: 2026-2027
EVIDENCE LEVEL	★★★★■ — Phase 2 completed with exceptional results; Phase 3 ongoing; Not yet FDA approved
REGULATORY STATUS	NOT YET FDA APPROVED. Breakthrough Therapy Designation granted by FDA (2023). Phase 3 TRIUMPH trials underway. Expected approval: 2026-2027 (if Phase 3 successful). NOT available through compounding pharmacies. Research chemical sources only — unverified quality.
CONTRAINDICATIONS	Personal/family history of medullary thyroid carcinoma (MTC) or MEN-2 syndrome, pregnancy, severe renal impairment, pancreatitis history, pre-existing tachycardia or arrhythmia (caution — GCGR effect)

How Retatrutide Compares to Other GLP-1 Class Drugs

Drug	Receptors	Max Weight Loss	Trial Duration	Status	Titration	Key Risk
Semaglutide (Wegovy)	GLP-1R only	~15% (2.4mg)	68 weeks	FDA Approved	16 weeks	Thyroid C-cell
Tirzepatide (Zepbound)	GLP-1R + GIP-R	~21% (15mg)	72 weeks	FDA Approved	20 weeks	Thyroid C-cell
Retatrutide	GLP-1R + GIP-R + GCGR	~24% (12mg)	24 weeks	Phase 3	20+ weeks	Tachycardia added
Cagrilintide + Sema (CagriSema)	GLP-1R + AMY1-3	~23% (combo)	32 weeks	Phase 3	20 weeks	GI side effects

★ Retatrutide row highlighted. Phase 3 data pending — current figures from Phase 2 trial.

Retatrutide Stacking — What to Combine, What to Avoid

■ COMBINE	**AOD-9604 (300-500mcg SC fasted AM)**
	Different mechanism — AOD activates beta-3 adrenergic receptors for visceral fat lipolysis. Retatrutide handles appetite/metabolic rate; AOD-9604 targets fat cells directly. Additive fat loss.
■ COMBINE	**Thymosin Alpha-1 (1.6mg SC 2x/week)**
	No interaction. GLP-1 class drugs don't affect the immune system. TA-1 can provide immune support and reduce GI inflammation that sometimes worsens GI side effects from high-dose triple agonists.
■ COMBINE	**BPC-157 (250mcg oral or SC)**
	BPC-157 may actually help manage the GI side effects of Retatrutide by repairing gut epithelium and reducing GI inflammation. Completely different mechanism. Highly complementary.
■ COMBINE	**GHK-Cu (topical or low-dose SC)**
	No interaction with GLP-1/GIP/GCGR axis. Can support skin quality during rapid weight loss (when loose skin becomes a concern). Compatible.
■ CAUTION	**Ipamorelin / CJC-1295**
	No direct interaction, but rapid weight loss from Retatrutide + GH secretagogue-driven muscle preservation = potentially conflicting metabolic signals. The two can complement each other (preserve lean mass while losing fat) but monitor fasting glucose closely — both affect insulin sensitivity.
■ CAUTION	**MK-677**
	MK-677 raises blood sugar; Retatrutide lowers it. Net effect unclear at high doses of both. Potential GI side effect amplification. If combining, start MK-677 at lowest dose (12.5mg). Monitor glucose weekly.
■ AVOID	**Semaglutide OR Tirzepatide**
	ABSOLUTE — same receptor class. Combining GLP-1 agonists provides no additional benefit and dramatically increases pancreatitis, thyroid tumor, and severe GI event risk. Use ONE GLP-1 class drug at a time.
■ AVOID	**Insulin (without physician oversight)**

	Compounding hypoglycemia risk. Retatrutide already lowers blood sugar significantly. Adding insulin without careful titration can cause dangerous hypoglycemia. Physician-managed only.
■ AVOID	**Any nitrate medication**
	Same rule as all GLP-1 class drugs — glucagon component adds cardiovascular activity. Nitrates + cardiovascular-active peptides = unpredictable blood pressure responses.

■ Important Note on Availability

Retatrutide is currently ONLY available through: (1) Eli Lilly Phase 3 clinical trial enrollment (TRIUMPH program — check clinicaltrials.gov), or (2) Unregulated research chemical vendors with unverified quality. It is NOT available through licensed compounding pharmacies and does NOT have FDA approval. Exercise extreme caution with any vendor claiming to sell it — purity and authenticity cannot be verified. For legitimate access, enroll in a clinical trial.

THE COMPLETE PRACTICAL

PEPTIDE STACKING & USAGE GUIDE

How to Use · What to Combine · What to Avoid · Cycle Protocols · Goal-Based Stacks

GOAL-BASED STACKS
8 Complete Protocols

COMBINATION GUIDE
Safe & Unsafe Pairings

STEP-BY-STEP PROTOCOLS
Timing · Dosing & Cycling

INTERACTION WARNINGS
What NOT To Mix

FOR EDUCATIONAL PURPOSES ONLY — Not Medical Advice — Consult a Licensed Healthcare Provider
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1 Before You Start

Critical Safety Rules — Read This First

■ This Guide is For Education Only

Everything in this guide is for research and educational purposes. Peptides (except FDA-approved ones) are not approved for human use. Always work with a licensed physician experienced in peptide therapy. Self-administration carries serious risks including contamination, incorrect dosing, and unknown long-term effects.

10 Rules Every User Must Know

1. Start Low, Go Slow	Never start at full dose. Begin at 25-50% of the standard dose for the first week. Your body needs time to adapt. Many side effects (nausea, water retention, headache) are dose-dependent and disappear when you titrate slowly.
2. One New Peptide at a Time	Never add multiple new peptides simultaneously. Introduce one new compound every 2-3 weeks. This way, if you experience a side effect, you know exactly which peptide caused it.
3. Source Matters Enormously	Research peptides are unregulated. Contamination, mislabeling, and incorrect dosing are common. Only use vendors with batch-specific HPLC + mass spectrometry COAs. Cheap peptides are not worth the risk.
4. Reconstitution & Storage	Always use bacteriostatic water (not sterile water) for multi-use vials. Reconstituted peptides must be refrigerated (4°C) and used within 30 days. Never freeze reconstituted peptides. Never shake — swirl gently.
5. Injection Technique	Use 29-31G insulin syringes. Rotate injection sites (left abdomen, right abdomen, left thigh, right thigh, back of arms). Pinch skin, inject at 45° angle. Wipe with alcohol before and after. Dispose sharps safely.
6. Cycle On and Off	Most peptides require cycling to prevent receptor desensitization. A common framework: 8-12 weeks ON, 4-6 weeks OFF. Some (BPC-157, TA-1) can be used for shorter acute cycles. GLP-1s are different — continuous use is standard.
7. Monitor Labs	Before starting GH-axis peptides (Ipamorelin, Sermorelin, CJC-1295, MK-677): get baseline IGF-1, fasting glucose, and HbA1c. Recheck at 8 weeks. Elevated IGF-1 or glucose signals you need to reduce dose or stop.
8. Peptides are Not Steroids	GH secretagogues work by stimulating your OWN pituitary — they will not work if your pituitary is suppressed or if you are already on exogenous HGH. They also take 4-12 weeks to show meaningful results. Patience is required.
9. Drug Interactions	Peptides can interact with: insulin (blood sugar risk), corticosteroids (reduce effectiveness of GH axis peptides), immunosuppressants (interact with immune peptides), blood pressure meds (PT-141 raises BP). Always disclose all medications to your physician.
10. Pregnancy & Cancer	Never use unapproved peptides if pregnant, breastfeeding, or if you have a history of cancer (especially hormone-sensitive cancers). GH-axis peptides and angiogenic peptides are particularly contraindicated.

2 How to Read a Peptide Protocol

Understanding Dosing, Timing & Cycling

Anatomy of a Peptide Protocol

Every peptide protocol in this guide follows the same format. Here is exactly how to read it:

ELEMENT	WHAT IT MEANS	EXAMPLE
Peptide Name	The compound being used	Ipamorelin
Dose	Amount per injection. mcg = micrograms; mg = milligrams. 1mg = 1,000mcg	200 mcg
Frequency	How many times per day/week. EOD = every other day	2x daily
Timing	When relative to food, sleep, or exercise. Fasted = no food 1-2h before/after	30 min pre-bed
Route	How it's administered: SC = subcutaneous; IM = intramuscular; IN = intranasal	SC
Cycle Length	How long to run the protocol before taking a break	12 weeks ON
Off Period	The break time to prevent receptor desensitization or tolerance	4 weeks OFF
Stack Partner	Another peptide specifically chosen to amplify effects	+ CJC-1295
Loading Phase	Initial higher-dose period to saturate receptors (some peptides)	Week 1: 500mcg
Maintenance Phase	Standard ongoing dose after loading	Week 2+: 300mcg
Evidence Level	★ = weak animal data only; ★★★★★ = FDA approved with RCTs	★★★■■

How the GH Axis Works — Why Timing Matters

Growth hormone secretagogues work by mimicking your body's natural hormone signals. The pituitary gland releases GH in **pulses** — primarily at night during deep sleep and after exercise. To maximize results:

Inject at bedtime (10-11pm) ★★★★★

GH is naturally highest 1-2 hours after falling asleep. Injecting just before bed amplifies this natural pulse — this is the most effective timing window.

Fast 2 hours before injection ★★★★■

Elevated insulin from food blunts GH release. An empty stomach maximizes the GH pulse you're stimulating. Even 100 calories of fat or protein can reduce response by 40-60%.

Pre-workout injection (optional 2nd dose) ★★★■■

A second injection 30 minutes before training capitalizes on exercise-induced GH release. Only do this if doing 2x daily dosing. Skip if blood sugar is a concern.

Avoid dosing after carbohydrate-rich meals

★★★★■

Insulin and GH are antagonistic. Post-meal GH secretagogue dosing is largely wasted. If you can only do one injection, bedtime fasted is always the right call.

3 Peptide Quick-Reference Card

All Major Peptides at a Glance

Peptide	Category	Dose	Timing	Route	Cycle	Combines With	Evidence
Ipamorelin	GH Secr.	200-300mcg	Bedtime (fasted)	SC	12 on/4 off	CJC-1295 no DAC	★★★■
CJC-1295 no DAC	GH Secr.	100-200mcg	With Ipamorelin	SC	12 on/4 off	Ipamorelin	★★★★
CJC-1295 DAC	GH Secr.	1-2mg	Once weekly	SC	12 on/4 off	Ipamorelin	★★★★
Sermorelin	GH Secr.	200-500mcg	Bedtime (fasted)	SC	12 on/4 off	Ipamorelin	★★★★
GHRP-6	GH Secr.	100-300mcg	Pre-meal or bedtime	SC	8 on/4 off	CJC-1295 no DAC	★★★
MK-677	GH Secr.	10-25mg	Evening (with food)	Oral	16+ wks on	Ipamorelin (separate)	★★★★
Tesamorelin	GH Secr.	2mg	Daily morning	SC	24+ wks	Ipamorelin	★★★★★
BPC-157	Healing	250-500mcg	Near injury or oral	SC/Oral	4-8 wks acute	TB-500	★★★
TB-500	Healing	2-2.5mg	2x/week	SC	4-8 wks	BPC-157	★★★
GHK-Cu	Healing/Skin	1-10mg	Systemic or topical	SC/Topical	Ongoing (topical)	BPC-157, TB-500	★★★
Semaglutide	GLP-1	0.25→2.4mg	Weekly any time	SC	Continuous	Tirzepatide (do not combine)	★★★★★
Tirzepatide	GLP-1	2.5→15mg	Weekly any time	SC	Continuous	Semaglutide (do not combine)	★★★★★
AOD-9604	Fat Loss	300-500mcg	Fasted AM	SC	12 on/4 off	GH secretagogues	★★★
Semax	Cognitive	100-600mcg	Morning	Intranasal	4-8 wks on/2 off	Selank	★★★★
Selank	Cognitive	250-500mcg	Morning or as needed	Intranasal	4-8 wks on/2 off	Semax	★★★★
Dihexa	Cognitive	10-30mg	Morning	Oral/Transdermal	4 wks on/4 off	Selank (caution)	★★
PT-141	Sexual	1.75mg	45 min before activity	SC	As needed (max 8x/mo)	Semax (separate)	★★★★
Thymosin Alpha-1	Immune	1.6mg	2x/week	SC	4-12 wks	BPC-157	★★★★
KPV	Immune/GI	200-500mcg	With meals (oral)	Oral/SC	4-8 wks	BPC-157	★★
Epitalon	Anti-aging	5-20mg	Evening	SC	10-20 day cycles, 1-2x/yr	MOTS-c, GHK-Cu	★★
MOTS-c	Anti-aging	5-10mg	Pre-workout or morning	SC	8-12 wks	Epitalon	★★

Humanin	Anti-aging	2-5mg	Morning	SC	8-12 wks	MOTS-c, Epitalon	★★
SS-31	Cardiovascular	0.1mg/kg	Morning	SC	8-12 wks	MOTS-c	★★★
Thymosin Beta-4/TB-500	Healing	2mg	2x/week	SC	4-8 wks	BPC-157	★★★
Oxytocin	Social/Mood	24-40 IU	30 min before event	Intranasal	As needed	Selank	★★★
Cerebrolysin	Neuro	5-30mL	Daily course	IV/IM	10-30 day cycles	Semax	★★★★

Key: SC = subcutaneous injection | IM = intramuscular | IN = intranasal | Evidence: ★ = animal only → ★★★★★ = FDA Approved RCTs

4 Goal-Based Stacking Protocols

Complete Protocols Organized by Your Goal

How to Use These Protocols

Each protocol below is a complete, ready-to-follow plan. The peptides listed have been chosen because they complement each other mechanistically — meaning they work on different but additive pathways. DO NOT combine protocols from different goals unless specifically noted. Run ONE goal protocol at a time.

■ A. Body Recomposition — Lean Muscle Gain + Fat Loss

PEPTIDES	STACK: Ipamorelin (200-300mcg SC) + CJC-1295 no DAC (100-200mcg SC) + BPC-157 (250mcg SC optional, injury support) BEGINNER: Ipamorelin only → Add CJC after 4 weeks	**DURATION**	12 weeks ON → 4-6 weeks OFF. Repeat up to 3 cycles/year.
TIMING	Both Ipamorelin + CJC-1295 together: 30-45 min pre-bed, fasted (2h post-meal). BPC-157: morning or near injury site.	**EXPECTED**	Week 4-6: improved recovery + sleep quality. Week 8-12: lean mass increase, reduced body fat, improved skin. Results accelerate with progressive resistance training.
WHY THEY WORK TOGETHER	WHY TOGETHER: Ipamorelin triggers GH pulse from pituitary. CJC-1295 (no DAC) amplifies and extends that pulse. Together they produce 3-5x more GH than either alone — without elevating cortisol (unlike GHRP-6). BPC-157 adds tendon/ligament resilience to support heavier training loads.	**WARNINGS**	Do NOT add MK-677 simultaneously (GH axis overload). Monitor IGF-1 and fasting glucose at 8 weeks. Avoid if diabetic.

■ B. Healing & Injury Recovery

PEPTIDES	STACK: BPC-157 (250-500mcg SC or oral) + TB-500 (2mg SC, 2x/week) ADD-ON for Inflammation: + KPV (500mcg oral) if gut/systemic inflammation component ADD-ON for Tissue Rebuilding: + GHK-Cu (2mg SC weekly)	**DURATION**	4-8 weeks for acute injury. Chronic: 8-12 weeks. BPC-157 can repeat indefinitely for GI conditions.
TIMING	BPC-157: Morning SC near injury site OR oral capsule for GI conditions. TB-500: 2x weekly SC (Monday/Thursday). GHK-Cu: weekly. KPV: with meals if oral.	**EXPECTED**	Week 1-2: reduced inflammation and pain. Week 3-4: accelerated tissue repair (fibroblast activation). Week 6-8: functional recovery and strength return.
WHY THEY WORK TOGETHER	WHY TOGETHER: BPC-157 activates local repair via VEGF/nitric oxide and fibroblast migration. TB-500 works systemically via actin remodeling and cell migration — different mechanisms that complement BPC-157 perfectly. GHK-Cu adds collagen synthesis. Together they cover systemic, local, and structural repair.	**WARNINGS**	Do NOT inject TB-500 directly into injury — inject SC nearby. BPC-157 near the injury is fine. Active cancer is a contraindication for both (angiogenesis concern).

■ C. Fat Loss — GLP-1 Protocol

PEPTIDES	OPTION 1 (BEST): Tirzepatide 2.5mg SC/week → Titrate up to 15mg over 6 months (requires Rx) OPTION 2: Semaglutide 0.25mg SC/week → Titrate to 2.4mg over 16 weeks (requires Rx) ADD-ON (Research): AOD-9604 300-500mcg SC fasted AM (visceral fat targeting, no Rx needed) DO NOT combine Semaglutide + Tirzepatide — same receptor class, no benefit, increased risk	**DURATION**	Continuous (GLP-1 drugs are designed for long-term use). AOD-9604: 12 weeks on/4 off.
TIMING	GLP-1: Same day weekly (e.g., every Sunday evening). AOD-9604: Every morning 30 min before breakfast.	**EXPECTED**	Month 1-2: 5-8% weight loss (GLP-1). Month 3-6: 12-20% body weight reduction (tirzepatide). AOD-9604 adds targeted visceral fat reduction.
WHY THEY WORK TOGETHER	WHY TOGETHER: GLP-1s suppress central appetite and slow gastric emptying. AOD-9604 directly activates fat cell beta-3 receptors for lipolysis without affecting insulin — completely different mechanism. No receptor competition, highly complementary.	**WARNINGS**	Titrate slowly to minimize nausea. Take with small protein-rich meal if nausea is severe. Monitor for muscle mass loss — add resistance training. Thyroid cancer history = absolute contraindication for GLP-1s.

■ D. Cognitive Enhancement & Neuroprotection

PEPTIDES	STACK: Semax (400-600mcg IN) + Selank (500mcg IN) ADVANCED ADD-ON: + Cerebrolysin (5-10mL IM, 10-day cycles) EXPERIMENTAL: + Dihexa (10mg oral daily) — use extreme caution, very limited human data	**DURATION**	Semax/Selank: 4-8 weeks on, 2-4 weeks off. Cerebrolysin: 10-30 day cycle, repeat quarterly. Dihexa: 4 weeks max, 4 weeks off.
TIMING	Semax: Morning intranasal (1-2 sprays per nostril). Selank: Morning or pre-stressful event intranasal. Cerebrolysin: IM injection daily for 10-30 day cycles. Dihexa: morning oral if using.	**EXPECTED**	Week 1-2: improved mood, anxiety reduction (Selank), mental clarity. Week 2-4: enhanced focus, working memory, faster information processing (Semax BDNF upregulation). Week 4-8: sustained cognitive improvements.
WHY THEY WORK TOGETHER	WHY TOGETHER: Semax elevates BDNF/NGF — growing new neural connections. Selank regulates anxiety/stress pathways that normally inhibit cognitive performance. They act on completely different systems and synergize powerfully. Cerebrolysin adds exogenous neurotrophic factors for accelerated repair and neurogenesis.	**WARNINGS**	Do NOT use Semax with stimulants (amphetamines, high-dose caffeine) — overstimulation risk. Dihexa is research-only with zero human safety data — use with extreme caution. Avoid Dihexa + Semax simultaneously.

■ E. Anti-Aging & Longevity Protocol

PEPTIDES	STACK: Epitalon (10mg SC daily × 10 days, 2x/year) + GHK-Cu (2-5mg SC weekly or topical daily) ADD-ON: MOTS-c (5-10mg SC weekly) + Humanin (2-5mg SC weekly) FOUNDATION: Ipamorelin + CJC-1295 (body recomposition stack, run as primary)	**DURATION**	Epitalon: 10-day on cycle, 2x per year (spring/fall). MOTS-c/Humanin: 12 weeks on, 8 weeks off. GHK-Cu: ongoing if topical.
TIMING	Epitalon: Evening SC daily for 10-day course only, then stop. GHK-Cu: Weekly SC injection OR daily topical serum. MOTS-c + Humanin: Weekly SC injections on different days.	**EXPECTED**	Months 1-3: improved sleep quality, skin texture, energy levels. 6-12 months: telomere length improvement (Epitalon cycles). Ongoing: mitochondrial efficiency (MOTS-c), reduced systemic inflammation.
WHY THEY WORK TOGETHER	WHY TOGETHER: Each targets a different hallmark of aging — Epitalon: telomere extension/epigenetics. GHK-Cu: gene expression + collagen repair. MOTS-c: mitochondrial biogenesis + insulin sensitivity. Humanin: neuronal protection against amyloid. No receptor overlap — all additive.	**WARNINGS**	Epitalon is theoretically contraindicated in active cancer (telomerase also active in tumor cells). MOTS-c/Humanin have limited human data — this is cutting-edge territory. Do NOT combine with exogenous HGH.

❤■ F. Sexual Health Protocol

PEPTIDES	WOMEN (HSDD): PT-141 1.75mg SC → 45 min before activity. Max 8x/month. MEN (Libido/ED): PT-141 1mg SC → Can combine with PDE5 inhibitor cautiously (monitor BP) ADD-ON for both: Selank 250mcg IN (reduces performance anxiety, enhances mood)	**DURATION**	As needed. Not a daily protocol. PT-141 max 8x per month. Tolerance develops with overuse — take breaks of at least 1 week between uses.
TIMING	PT-141: SC injection 45-90 minutes before desired activity. Selank: 15-30 min before activity or event. Do NOT take on a full stomach (nausea risk).	**EXPECTED**	PT-141: 60-80% response rate for enhanced desire and arousal within 90 minutes. Selank: reduced anxiety allowing PT-141 to work more effectively.
WHY THEY WORK TOGETHER	WHY TOGETHER: PT-141 activates central MC4R receptors for desire. Selank reduces GABAergic anxiety that often blocks libido. They operate on completely different brain circuits with additive effect.	**WARNINGS**	PT-141 raises blood pressure 5-10mmHg. Do NOT combine with nitrates (nitroglycerin, amyl nitrite) — dangerous hypotension. Nausea is common — start at 1mg first use. Chronic use causes hyperpigmentation of moles.

■ G. Immune Support & Recovery Protocol

PEPTIDES	STACK: Thymosin Alpha-1 (1.6mg SC, 2x/week) + BPC-157 (250-500mcg SC or oral) ADD-ON for GI-specific inflammation: + KPV (500mcg oral with meals) ACUTE ILLNESS (cold/flu/post-COVID): TA-1 every day × 5-7 days, then 2x/week	**DURATION**	Baseline immune support: 4-8 weeks. Post-illness recovery: 2-4 weeks intensive then 4-6 weeks maintenance.
TIMING	TA-1: Monday and Thursday SC. BPC-157: Daily oral or SC (different site). KPV: 3x daily with meals.	**EXPECTED**	Week 1-2: reduced sick duration, improved energy. Week 2-4: normalized inflammatory markers. Week 4-8: restored immune cell function, improved gut barrier integrity.
WHY THEY WORK TOGETHER	WHY TOGETHER: TA-1 acts on the adaptive immune system (T-cells, NK cells, dendritic cells). BPC-157 repairs the mucosal barrier and reduces systemic inflammation via different pathways. KPV targets intestinal NF-kB for local gut inflammation. Three different but complementary mechanisms.	**WARNINGS**	TA-1 is generally extremely safe. Caution in active autoimmune flares — can upregulate immune activity that may worsen flares. Stop if autoimmune symptoms worsen.

■ H. Sleep Optimization Protocol

PEPTIDES	STACK: Ipamorelin (200mcg SC) + DSIP (Delta Sleep Inducing Peptide, 200-600mcg SC) ALTERNATIVE (oral only): MK-677 (12.5-25mg oral at bedtime) ADD-ON: Epitalon (during active cycle) supports melatonin regulation	**DURATION**	MK-677: 16+ weeks continuous (builds up over time). Ipamorelin/DSIP: 12 weeks on, 4 off.
TIMING	Ipamorelin + DSIP: SC 30 minutes before bed, fully fasted (2h post-meal). MK-677: oral with small snack at bedtime (reduces nausea).	**EXPECTED**	Week 1-4: improved sleep latency and REM quality. Week 4-8: increased slow-wave (deep) sleep — this is where GH release and cellular repair happen. Week 8-12: improved body composition, cognitive sharpness from better sleep.
WHY THEY WORK TOGETHER	WHY TOGETHER: Ipamorelin triggers the natural nocturnal GH pulse. DSIP specifically promotes slow-wave sleep architecture. MK-677 independently increases REM and slow-wave sleep. They target different sleep mechanisms with additive benefits.	**WARNINGS**	MK-677 causes water retention and may worsen insulin resistance — avoid if diabetic. DSIP has limited human data — consider it experimental. Morning grogginess is common with MK-677 first 2-4 weeks.

5 Master Compatibility Matrix

What Combines With What — At a Glance

How to Read This Matrix

■ COMBINE = These peptides work well together and are commonly stacked. ■ CAUTION = Can combine but monitor closely / start low / not for beginners. ■ DO NOT COMBINE = Either redundant (same receptor), dangerous interaction, or unknown risk. — = No meaningful interaction; can use in same protocol without issue.

GH Secretagogue Compatibility

	Ipamorelin	CJC no DAC	CJC DAC	Sermorelin	GHRP-6	MK-677	Tesamorelin
Ipamorelin	—	■ BEST COMBO	■ GOOD	■ REDUNDANT	■ SKIP	■ MONITOR IGF-1	■ SKIP
CJC no DAC	■ BEST COMBO	—	■ DO NOT COMBINE	■ REDUNDANT	■ OK	■ MONITOR	■ SKIP
CJC DAC	■ GOOD	■ DO NOT COMBINE	—	■ SKIP	■ OK	■ MONITOR	■ SKIP
GHRP-6	■ SKIP	■ OK	■ OK	■	—	■ SKIP	■ SKIP
MK-677	■ MONITOR	■ MONITOR	■ MONITOR	■	■ SKIP	—	■ SKIP

Healing & Immune Compatibility

	BPC-157	TB-500	GHK-Cu	TA-1	KPV	Ipamorelin	Semaglutide
BPC-157	—	■ BEST	■ GREAT	■ GREAT	■ GREAT	■ OK	■ OK
TB-500	■ BEST	—	■ OK	■ OK	■ REDUNDANT	■ OK	■ OK
GHK-Cu	■ GREAT	■ OK	—	■ OK	■ OK	■ OK	■ OK
TA-1	■ GREAT	■ OK	■ OK	—	■ GREAT	■ OK	■ OK
KPV	■ GREAT	■	■ OK	■ GREAT	—	■ OK	■ OK

Cognitive Peptide Compatibility

	Semax	Selank	Dihexa	Cerebrolysin	Oxytocin	BPC-157	MOTS-c
Semax	—	■ BEST	■ AVOID	■ GREAT	■ MONITOR	■ OK	■ OK
Selank	■ BEST	—	■ CAUTION	■ OK	■ GREAT	■ OK	■ OK
Dihexa	■ AVOID	■ CAUTION	—	■ CAUTION	■	■ OK	■ OK
Cerebrolysin	■ GREAT	■ OK	■ CAUTION	—	■	■ OK	■ OK

Legend: ■ = Combine freely | ■ = Use caution / monitor | ■ = Do not combine | — = Same peptide (self)

6 Dangerous Combinations

What NEVER to Mix — And Why

■ These combinations can cause serious harm or are simply wasteful

Some combinations cause real physiological harm. Others are just redundant (wasted money, no benefit, increased side effects). Both types are listed here so you know exactly what to avoid.

■ NEVER	**CJC-1295 no DAC + CJC-1295 DAC**	*DANGEROUS — Receptor Saturation + Wasted Money*

These are two versions of the same peptide targeting the same GHRH receptors. Using both simultaneously saturates all available receptors without additional benefit. You get the side effects of both (water retention, insulin resistance) with no additional GH response. NEVER use both. Pick one format — no DAC for natural pulsatile release, DAC for once-weekly convenience.

■ NEVER	**Semaglutide + Tirzepatide**	*DANGEROUS — Same Receptor Pathway, Double Risk*

Both are GLP-1 receptor agonists. Combining them does not produce double weight loss — it produces double nausea, double pancreatitis risk, double thyroid risk, and double the chance of severe GI events. There is no clinical benefit. This is an absolute contraindication. Use ONE GLP-1 at a time.

■ NEVER	**Any GH Secretagogue + Exogenous HGH**	*REDUNDANT + Costly + Risk of Acromegaly*

GH secretagogues (Ipamorelin, CJC-1295, Sermorelin) work by telling your pituitary to make MORE of your own GH. If you're already injecting exogenous HGH, your pituitary is suppressed via negative feedback. The secretagogues have no pituitary to stimulate — they do nothing. Plus, combining may lead to excess IGF-1, joint pain, carpal tunnel syndrome, and long-term acromegaly risk.

■ NEVER	**PT-141 + Nitrates (nitroglycerin, amyl nitrite/poppers)**	*LIFE-THREATENING — Severe Hypotension*

PT-141 raises blood pressure 5-10 mmHg. Nitrates cause vasodilation and lower blood pressure. Together they can cause catastrophic hypotensive collapse. This is not a theoretical risk — it is a documented life-threatening interaction. PT-141 combined with nitrates is absolutely contraindicated.

■ NEVER	**GHRP-6 + Ipamorelin simultaneously**	*LARGELY REDUNDANT — Cortisol Risk*

Both are GHS-R1a agonists. GHRP-6 causes significant cortisol and prolactin elevation — Ipamorelin was specifically designed to fix this problem (selective GHS with no cortisol/prolactin effect). Combining them cancels out Ipamorelin's key advantage. The cortisol elevation from GHRP-6 will blunt the clean GH release from Ipamorelin. Just use Ipamorelin.

■ NEVER	**Semax + Dihexa simultaneously**	*UNKNOWN DANGER — Neurological Overstimulation Risk*

Semax massively elevates BDNF and NGF. Dihexa potentiates the HGF/c-Met pathway leading to rapid synaptogenesis. Both together create uncontrolled neuronal growth signaling via completely different mechanisms. In animal studies, excessive BDNF combined with rapid synaptogenesis has caused seizure-like activity. No human safety data exists for this combination. Never use simultaneously.

■ CAUTION	**MK-677 + Any GH Secretagogue**	*HIGH IGF-1 + Insulin Resistance Risk*

MK-677 produces sustained, continuous GH/IGF-1 elevation. Adding pulsatile GH secretagogues (Ipamorelin, CJC-1295) on top creates chronically elevated IGF-1 which increases cancer risk, insulin resistance, and joint/tissue issues. If using both, use MK-677 at LOW dose (12.5mg) and monitor IGF-1 monthly. Not recommended for beginners.

■ CAUTION	**BPC-157/TB-500 + Active Cancer**	*Theoretical Angiogenesis Risk*

BPC-157 and TB-500 both promote angiogenesis (new blood vessel formation) — a core mechanism of their healing effects. This same mechanism theoretically feeds tumor vascularity in cancer patients. Neither has been proven to cause or accelerate cancer in humans, but the theoretical risk means you should absolutely avoid these in anyone with active malignancy or recent cancer history.

■ CAUTION	**GLP-1 Agonists + Insulin**	*Blood Sugar Collapse Risk*

GLP-1 agonists already reduce blood sugar significantly. Adding insulin creates compounding hypoglycemia risk. If diabetic and using both, this MUST be managed by a physician with careful glucose monitoring. Spontaneous hypoglycemia can be life-threatening. Never adjust insulin independently when starting a GLP-1.

■ CAUTION	**PT-141 + PDE5 Inhibitors (Viagra/Cialis)**	*Blood Pressure Monitoring Required*

PT-141 raises blood pressure 5-10 mmHg. PDE5 inhibitors lower it via vasodilation. The effects may partially counteract, but the net hemodynamic effect is unpredictable. This combination is used by some physicians for refractory ED, but requires BP monitoring. Not for men with cardiovascular disease or hypertension.

7 Individual Peptide How-To Cards

Step-by-Step Usage Guide for Each Peptide

How to Use These Cards

Each card below tells you exactly: what to use before this peptide (prep), what to use WITH it (stack), what NOT to use with it (avoid), the injection protocol, and what to expect week by week.

GH Secretagogue — Safest GHRP

BEFORE YOU START	Get baseline labs: fasting glucose, IGF-1, HbA1c. Do NOT use if on exogenous HGH. Ensure you have bacteriostatic water for reconstitution. Confirm no active cancer.
RECONSTITUTION	Add 2mL bacteriostatic water to 5mg vial → 2,500mcg/mL. Each 0.1mL = 250mcg.
PROTOCOL	200-300mcg SC, 30-45 min before bed. FASTED (nothing to eat 2h prior, nothing until morning). Optional 2nd dose: 30 min pre-workout on training days. Rotate sites: abdomen left/right.
■ COMBINE WITH	CJC-1295 no DAC (inject at same time, same site is fine). BPC-157 (unrelated, can run simultaneously).
■ AVOID	GHRP-6 (cancels Ipamorelin's cortisol advantage). MK-677 (IGF-1 overload risk — monitor if combining). CJC DAC (use one CJC, not both).
EXPECT — WEEK 1	Improved sleep depth. Some notice faster sleep onset. Mild water retention in hands (normal).
EXPECT — WEEK 4	Noticeable body composition changes begin. Improved skin texture. Faster gym recovery.
EXPECT — WEEK 12+	Lean mass gain, reduced body fat, improved energy. IGF-1 elevated 30-60% above baseline.
CYCLING	12 weeks ON → 4-6 weeks OFF. After 3 cycles, take 3 months off.
MONITORING	IGF-1 and fasting glucose at Week 8. Stop if IGF-1 >300 ng/mL or fasting glucose >100 mg/dL.

TH Ipamorelin

GH Secretagogue — GHRH Analog

BEFORE YOU START	Always pair with Ipamorelin. Using alone is suboptimal — CJC amplifies the GH pulse, but Ipamorelin provides the initial trigger. Baseline labs same as Ipamorelin.
RECONSTITUTION	Add 2mL bacteriostatic water to 2mg vial → 1,000mcg/mL. Each 0.1mL = 100mcg.
PROTOCOL	100-200mcg SC, injected at EXACTLY the same time as Ipamorelin (bedtime, fasted). Can draw both into same syringe or inject separately — either works.
■ COMBINE WITH	Ipamorelin (mandatory partner). BPC-157 (add-on, separate injection).
■ AVOID	CJC-1295 DAC (never use both CJC versions simultaneously — total receptor overload). Sermorelin (same receptor class — redundant).
EXPECT — WEEK 1	Same as Ipamorelin alone but more pronounced — deeper sleep, more vivid dreams (GH effect).
EXPECT — WEEK 4	Body recomposition effects stronger than Ipamorelin alone. Better muscle fullness.
EXPECT — WEEK 12+	Superior lean mass and fat loss vs. either peptide alone. SYNERGY is the reason for combining.
CYCLING	12 weeks ON → 4-6 weeks OFF. Match exactly to Ipamorelin cycle.
MONITORING	Same as Ipamorelin — IGF-1 and fasting glucose at Week 8.

Healing & Recovery — Tissue Repair

BEFORE YOU START	No special labs required. Identify the injury site for targeted injection. For GI conditions: oral capsule form (250-500mcg in 00 capsules) is superior to injections.
RECONSTITUTION	Add 2mL bacteriostatic water to 5mg vial → 2,500mcg/mL. Each 0.1mL = 250mcg.
PROTOCOL	INJURY: 250-500mcg SC daily, inject near (not INTO) the injured tissue. GI CONDITIONS: 250-500mcg oral (capsule or dissolved in water), fasted. SYSTEMIC: 250-500mcg SC abdomen daily.
■ COMBINE WITH	TB-500 (best combination — different but complementary mechanisms). Thymosin Alpha-1 (for immune+recovery). GHK-Cu (for additional tissue repair).
■ AVOID	Active cancer (angiogenesis concern). Do not inject directly into joints — near the joint only.
EXPECT — WEEK 1	Reduced local inflammation and pain within 3-7 days. Improved gut symptoms in days if oral.
CYCLING	Acute injury: 4-8 weeks, stop when healed. GI: 4-6 weeks. Can repeat as needed.
MONITORING	No specific labs needed. Monitor for unusual mole changes (angiogenesis concern — very low risk).

gment)

Healing & Recovery — Systemic Repair

BEFORE YOU START	TB-500 is systemic — it travels through the bloodstream to all tissues. Best combined with BPC-157 for comprehensive coverage. No special labs needed.
RECONSTITUTION	Add 2mL bacteriostatic water to 5mg vial → 2,500mcg/mL.
PROTOCOL	LOADING PHASE (Weeks 1-4): 2-2.5mg SC twice weekly (e.g., Mon/Thu). MAINTENANCE (Weeks 5-8): 1-2mg SC once weekly.
■ COMBINE WITH	BPC-157 (essential partner). GHK-Cu. Ipamorelin/CJC (for athletes — recovery + recomposition).
■ AVOID	Active malignancy. Do not use at full dose for extended periods without the loading/maintenance split.
EXPECT — WEEK 1	Systemic inflammation reduction. Athletes notice faster DOMS recovery.
CYCLING	4-8 weeks acute protocol. 3-4 month break between full cycles.
MONITORING	No required labs. Monitor for any unusual mole changes.

c)

GLP-1 Agonist — Fat Loss / Metabolic (REQUIRES Rx)

BEFORE YOU START	REQUIRES PRESCRIPTION. Baseline labs: HbA1c, fasting glucose, lipids, thyroid panel. Confirm no personal/family history of medullary thyroid carcinoma or MEN-2.
RECONSTITUTION	Pre-filled pens — no reconstitution needed. Auto-inject pen from pharmacy.
PROTOCOL	WEEK 1-4: 0.25mg SC weekly. WEEK 5-8: 0.5mg. WEEK 9-12: 1mg. WEEK 13-16: 1.7mg. WEEK 17+: 2.4mg (full Wegovy dose). NEVER skip titration — nausea will be severe.
■ COMBINE WITH	AOD-9604 (complementary fat loss — different mechanism). Resistance training mandatory to preserve muscle.
■ AVOID	Tirzepatide (same class — never combine). Insulin (hypoglycemia risk — physician managed only). Fatty/fried foods (amplify nausea dramatically).
EXPECT — WEEK 4	Appetite suppression. 2-4% weight loss. Nausea most common here — titrate slowly.
EXPECT — WEEK 12+	8-12% body weight lost. Significant metabolic improvement. Blood sugar control improving.
EXPECT — 6 MONTHS	12-18% body weight reduction. Full metabolic benefits. Some begin plateauing.
CYCLING	Continuous — GLP-1 drugs are designed for long-term use. Stopping causes weight regain.
MONITORING	HbA1c and lipids at 3 and 6 months. Report any pancreatitis symptoms immediately.

	Cognitive / Neuroprotective
BEFORE YOU START	No special labs required. Begin with lowest dose (100mcg) to assess response. Avoid if on psychiatric medications without physician guidance.
RECONSTITUTION	Typically comes as nasal spray (0.1% or 1%). Store refrigerated. 1% solution: 1 spray = ~100-150mcg. Adjust based on your bottle concentration.
PROTOCOL	BEGINNER: 100-200mcg IN (1-2 sprays) each morning for 1 week. STANDARD: 400-600mcg IN each morning (2-4 sprays per nostril). ACUTE COGNITIVE TASK: 600-900mcg 20 min before demanding work.
■ COMBINE WITH	Selank (perfect partner — Semax for cognitive activation, Selank for anxiety reduction). Cerebrolysin (powerful combination for neuroregeneration — run courses separately).
■ AVOID	Dihexa (combined synaptogenesis risk — overstimulation). Psychostimulants (amphetamines, high caffeine). Do not use for more than 8 consecutive weeks without a break.
EXPECT — WEEK 1	Mental clarity, motivation increase, slightly enhanced mood. Some feel mild headache first 2 days.
EXPECT — WEEK 2-4	Sustained cognitive improvements — faster processing, better working memory, improved focus.
EXPECT — WEEK 4-8	BDNF upregulation showing — enhanced neuroplasticity, improved learning capacity.
CYCLING	4-8 weeks ON → 2-4 weeks OFF. Tolerance develops with continuous use.
MONITORING	No labs required. Monitor for anxiety at higher doses — reduce dose if this occurs.

	Anxiolytic / Cognitive
BEFORE YOU START	No special labs required. Excellent first peptide for beginners — very forgiving safety profile. Note: this is the anxiolytic counterpart to Semax. Often used together.
RECONSTITUTION	Usually pre-formulated as nasal spray (0.15%). 1 spray = ~100mcg.
PROTOCOL	ANXIETY RELIEF: 250-500mcg IN as needed during stressful periods. DAILY PROTOCOL: 250-500mcg IN morning daily for 4-8 weeks. COMBINED WITH SEMAX: Semax in AM, Selank can be taken at same time or separately.
■ COMBINE WITH	Semax (perfect match — add cognitive power without adding anxiety). PT-141 (reduces performance anxiety). Oxytocin (social anxiety).
■ AVOID	Benzodiazepines (additive CNS depression — not dangerous but sedating). Heavy alcohol use during protocol.
EXPECT — WEEK 1	Reduced background anxiety. Calmer, more measured stress responses. Slight mood elevation.
EXPECT — WEEK 2-4	Improved emotional regulation. Better sleep quality. Enhanced social confidence.
EXPECT — WEEK 4-8	Sustained anxiolytic effect. BDNF upregulation improving resilience.
CYCLING	4-8 weeks ON → 2-4 weeks OFF. Can also be used as-needed for acute anxiety.
MONITORING	None required. Very safe compound.

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Sexual Health — MC4R Agonist (FDA Approved)

BEFORE YOU START	Measure baseline blood pressure. Ensure no nitrate medications being used. Eat a light meal 2-3 hours before (not fasted, but not full). Check moles/skin before starting.
RECONSTITUTION	Comes as pre-measured pen or vial (1.75mg/0.3mL). Keep refrigerated.
PROTOCOL	FIRST USE: Try 1mg SC (reduce from full 1.75mg dose) to assess nausea response. STANDARD: 1.75mg SC, 45-90 minutes before sexual activity. Inject in abdomen or thigh — NOT arm (too fast absorption). Lie down if nauseous.
■ COMBINE WITH	Selank (30 min before — reduces performance anxiety so PT-141 works better). Oxytocin 24-40 IU IN (enhances emotional bonding component).
■ AVOID	NITRATES — ABSOLUTE CONTRAINDICATION (life-threatening hypotension). Antihypertensives — monitor BP closely. Use max 8x per month.
EXPECTED EFFECTS	Within 45-90 min: increased desire, arousal, sensitivity. Erection firmness improves. Duration: 6-12 hours of enhanced responsiveness.
CYCLING	As-needed only. Maximum 8 uses per month. Take at least 1 week between consecutive uses.
MONITORING	Check BP 45 min after injection. Monitor moles for darkening/changes over months.

GH Secretagogue — Oral (Non-Peptide)

BEFORE YOU START	Baseline fasting glucose and HbA1c MANDATORY. MK-677 raises blood sugar. Do NOT use if diabetic, pre-diabetic, or insulin resistant without physician supervision.
RECONSTITUTION	Comes as oral capsules or liquid. Standard: 12.5mg or 25mg capsules. No injection needed.
PROTOCOL	START: 12.5mg oral each evening with a small snack (reduces nausea/lethargy). AFTER 4 WEEKS: Increase to 25mg if tolerating well. Always take in the EVENING — GH pulse timing. Morning use disrupts sleep architecture.
■ COMBINE WITH	Ipamorelin (separate timing — MK-677 evening, Ipamorelin bedtime). BPC-157 (healing support during recomposition). Thymosin Alpha-1 (immune support).
■ AVOID	Insulin/antidiabetics without physician oversight. Do NOT combine with CJC-1295 DAC (sustained + sustained = chronic IGF-1 elevation — high risk).
EXPECT — WEEK 12+	Peak effects. IGF-1 elevated 40-80%. Skin, hair, nail quality improving.
EXPECT — WEEK 4-8	Meaningful body composition changes — muscle fullness, fat reduction beginning.
CYCLING	MK-677 effects BUILD over time. Minimum effective cycle: 16-24 weeks. 4-month break after.
MONITORING	Fasting glucose every 4 weeks. HbA1c at end of cycle. Stop if fasting glucose >110 mg/dL.

Immune Modulation	
BEFORE YOU START	No special labs required. Consider baseline CBC with differential to track immune response. Ideal timing: before expected immune challenge (seasonal illness, travel, post-illness recovery).
RECONSTITUTION	1.6mg vial. Add 1mL bacteriostatic water → 1,600mcg/mL = full dose per injection.
PROTOCOL	STANDARD: 1.6mg SC Monday and Thursday (2x weekly). ACUTE ILLNESS/INTENSIVE: 1.6mg SC daily for 5-7 days, then drop to 2x weekly. MAINTENANCE: 1.6mg SC once weekly for ongoing immune support.
■ COMBINE WITH	BPC-157 (comprehensive recovery — immune + tissue repair). KPV (gut inflammation + immune). GHK-Cu (cellular regeneration).
■ AVOID	Active autoimmune disease flares — TA-1 boosts immune activity which may worsen autoimmune conditions. Not contraindicated in stable autoimmune disease, but monitor.
EXPECT — WEEK 2-4	Normalized immune cell ratios. Reduced chronic infections.
EXPECT — WEEK 4-8	Full immune system restoration. Improved response to vaccines and pathogens.
EXPECT — WEEK 1-2	Reduced sick duration if used during illness. Energy improvement.
CYCLING	4-12 weeks for immune restoration. Annual 4-8 week course for maintenance.
MONITORING	CBC with differential at start and end of cycle. Watch for autoimmune symptom flares.

Anti-Aging / Telomere Extension	
BEFORE YOU START	Consider getting telomere length tested (expensive but available through specialty labs). Confirm no active cancer diagnosis. Epitalon activates telomerase — the same enzyme active in cancer cells.
RECONSTITUTION	10mg vial. Add 2mL bacteriostatic water → 5,000mcg/mL. Dose 5-20mg daily.
PROTOCOL	STANDARD COURSE: 10mg SC daily for 10-20 consecutive days. Then STOP completely. Run this 10-20 day course 2x per year (spring and fall). Do NOT run continuously. Best taken in the evening (melatonin regulation).
■ COMBINE WITH	GHK-Cu (comprehensive anti-aging — different mechanisms). MOTS-c (mitochondrial + telomere protocol). Ipamorelin/CJC-1295 (GH axis + longevity).
■ AVOID	Active cancer or recent cancer history (absolute contraindication). Do NOT use continuously — pulsed cycles only. Do not combine with cancer treatments.
EXPECTED EFFECTS	Course 1: Improved sleep quality, melatonin normalization, subtle energy increase. After 2-3 courses over 1 year: measurable telomere length increase (per research). Long-term (years): reduced biological aging markers.
CYCLING	10-20 day course ONLY. 2x per year maximum. Never daily continuous use.
MONITORING	Telomere length testing (baseline and annually). Blood work including CBC and cancer markers.

Mitochondrial Peptide / Anti-Aging

BEFORE YOU START	Baseline fasting glucose and insulin sensitivity markers (HOMA-IR). MOTS-c works like a metabolic exercise mimetic — it activates AMPK. Best results when combined with actual exercise.
RECONSTITUTION	Add 2mL bacteriostatic water to 5mg vial → 2,500mcg/mL.
PROTOCOL	5-10mg SC weekly. Best injected pre-workout or morning on rest days. Some protocols use 2-3x weekly at lower dose (2-3mg).
■ COMBINE WITH	Epitalon (anti-aging protocol — run during/after Epitalon cycle). Humanin (complementary mitochondrial peptides). BPC-157 (metabolic + recovery).
■ AVOID	No known dangerous combinations. Limited human data — use with caution generally.
EXPECT — WEEK 4	Improved exercise performance and endurance. Better insulin sensitivity.
EXPECT — WEEK 8-12	Meaningful metabolic improvements. Reduced visceral fat (especially with exercise).
CYCLING	8-12 weeks ON → 8 weeks OFF. This is cutting-edge — conservative cycling recommended.
MONITORING	Fasting glucose and HOMA-IR at start and end of cycle.

8 Timing & Injection Schedule Templates

Weekly Calendars for Common Protocols

Schedule A — Body Recomposition (Ipamorelin + CJC-1295)

MON	TUE	WED	THU	FRI	SAT	SUN
Ipam 250mcg + CJC 150mcg Bedtime SC	Rest (no injection)	Ipam 250mcg + CJC 150mcg Bedtime SC	Rest	Ipam 250mcg + CJC 150mcg Bedtime SC	Rest or Ipam only	Rest
(bedtime, fasted)		(bedtime, fasted)		(bedtime, fasted)		

Schedule B — Healing & Recovery (BPC-157 + TB-500)

MON	TUE	WED	THU	FRI	SAT	SUN
BPC-157 250mcg SC (near injury)	BPC-157 250mcg SC	TB-500 2mg SC + BPC-157	BPC-157 250mcg SC	BPC-157 250mcg SC	TB-500 2mg SC (maintenance)	Rest
Daily BPC-157 morning		TB-500 Mon/Thu				

Schedule C — Immune Protocol (Thymosin Alpha-1 + BPC-157)

MON	TUE	WED	THU	FRI	SAT	SUN
TA-1 1.6mg SC	BPC-157 250mcg SC	BPC-157 250mcg SC	TA-1 1.6mg SC	BPC-157 250mcg SC	BPC-157 250mcg SC	Rest
Morning	Morning	Morning	Morning	Morning	Morning	

Schedule D — Anti-Aging (Epitalon 10-Day Cycle + GHK-Cu Weekly)

MON	TUE	WED	THU	FRI	SAT	SUN
Epitalon 10mg SC	Epitalon 10mg SC	Epitalon 10mg SC	Epitalon 10mg SC + GHK-Cu 2mg SC	Epitalon 10mg SC	Epitalon 10mg SC	Epitalon 10mg SC
Days 1-10 then STOP for 6+ months			GHK-Cu continues weekly			

Schedule E — Cognitive Stack (Semax + Selank — Intranasal)

MON	TUE	WED	THU	FRI	SAT	SUN
Semax 400mcg IN + Selank 500mcg IN	Semax 400mcg IN + Selank 500mcg IN	Semax 400mcg IN + Selank 500mcg IN	Semax 400mcg IN + Selank 500mcg IN	Semax 400mcg IN + Selank 500mcg IN	REST (or Selank only)	REST

Morning both	Morning	Morning	Morning	Morning		

9 Cycling Guide

On/Off Protocols — Preventing Tolerance & Receptor Downregulation

Why Cycling Matters

Peptide receptors respond to continuous stimulation by **downregulating** — reducing their sensitivity to the signal. This is the body's natural homeostasis. If you don't cycle, you'll need progressively higher doses to get the same effect, and when you stop, your baseline may be lower than when you started. Cycling preserves receptor sensitivity and maintains your body's natural hormone rhythms.

Peptide/Class	Typical ON	Typical OFF	Why	Notes
Ipamorelin	12 weeks	4-6 weeks	GHS-R1a desensitization	Match off-period to CJC-1295 partner
CJC-1295 no DAC	12 weeks	4-6 weeks	GHRH receptor downregulation	Always cycle with Ipamorelin
CJC-1295 DAC	12 weeks	6-8 weeks	Longer DAC half-life needs longer washout	Albumin-bound — longer off period needed
MK-677	16-24 weeks	8-12 weeks	Tolerance + insulin resistance	Longer cycle because effects build slowly
Sermorelin	12 weeks	4-6 weeks	Same as Ipamorelin	Gentler GH response — 3 cycles/year fine
BPC-157 (injury)	4-8 weeks (acute)	Stop when healed	Use until injury resolved, then stop	For chronic GI: can repeat every 3 months
TB-500	4-8 weeks	8-12 weeks	Allow natural repair to consolidate	Loading + maintenance protocol required
GHK-Cu (topical)	Ongoing	Monthly break (1 wk/month)	Copper accumulation (minimal risk)	Systemic SC: 8 wks on / 4 off
Semax	4-8 weeks	2-4 weeks	Neuroplasticity normalization	Tolerance develops faster at high doses
Selank	4-8 weeks	2-4 weeks	GABA pathway adaptation	Can also use as-needed without cycling
Thymosin Alpha-1	4-12 weeks	4-8 weeks	Immune system normalization	Acute illness = 1-2 weeks, then stop
Epitalon	10-20 days	6 months	Telomerase pulse activation — not continuous	Pulsed protocol critical — NEVER continuous
MOTS-c	8-12 weeks	8 weeks	AMPK pathway adaptation	Limited human data — conservative approach
GLP-1 (Sema/Tirz)	Continuous	N/A	Designed for long-term continuous use	Weight regain occurs with cessation
PT-141	As needed	1 week between uses	MC4R desensitization with overuse	Max 8x per month

10 Beginner vs. Advanced Protocols

Start Here If You're New to Peptides

Rule #1 for Beginners: ONE Peptide at a Time

Do NOT start with a stack. Run a SINGLE peptide for 4 weeks first to learn how your body responds. Once you know you tolerate it well (no unexpected side effects), you may add a second peptide. Advanced users ONLY stack after multiple solo cycles.

First-time peptide user. No prior experience.

Peptide	Ipamorelin ONLY
Dose	200mcg SC
Timing	30 min before bed, fasted
Cycle	8 weeks on, 4 weeks off
Goal	Learn response, improve sleep, begin body recomposition
DO NOT ADD	Anything else for the first 8 weeks
Labs	Baseline IGF-1 + fasting glucose → recheck at Week 8

Completed 1-2 solo Ipamorelin cycles. Good understanding of own respon

Peptides	Ipamorelin 200mcg + CJC-1295 no DAC 150mcg
Timing	Both SC bedtime together, fasted
Cycle	12 weeks on, 4-6 weeks off
Goal	Full body recomposition protocol
Optional Add	BPC-157 250mcg SC daily if dealing with any injuries
Labs	IGF-1 + fasting glucose at 8 weeks
Progress	If all good, add 3rd peptide in next cycle

Multiple cycles completed. Full understanding of body's response.

Option A (Recomp+Healing)	Ipamorelin + CJC-1295 + BPC-157 + TB-500
Option B (Recomp+Cognitive)	Ipamorelin + CJC-1295 + Semax + Selank (separate AM intranasal)
Option C (Full Stack)	Ipamorelin + CJC-1295 + BPC-157 + TA-1 + GHK-Cu
Rule	Never run more than 4-5 peptides simultaneously
Labs	Full panel: IGF-1, fasting glucose, HbA1c, CBC, CMP every 8 weeks
Caution	Consider physician supervision at this level

11 Frequently Asked Questions

The Most Common Questions — Answered Directly

Q: Do I need to inject every day?

A: For most GH secretagogues (Ipamorelin, CJC-1295): ideally yes, daily at bedtime for maximum results. However, even 5x per week produces strong results. If injecting daily feels overwhelming, starting with 5x/week (Mon-Fri) and rest weekends is perfectly effective. TB-500: only 2x weekly. Epitalon: daily for 10 days only, then a 6-month break.

Q: Can I miss a dose? What happens?

A: Missing 1-2 doses is not a problem. These peptides work by accumulating effects over weeks, not by a single injection. Do NOT double-dose to 'make up' for a missed injection — this doesn't help and increases side effects. Just resume your normal schedule the next day.

Q: I feel nothing after 2 weeks — is it working?

A: Yes, probably. Peptides like Ipamorelin/CJC work over 4-12 weeks of accumulation. The most common early sign is improved SLEEP QUALITY — you fall asleep faster and sleep deeper. Body composition changes take 6-10 weeks. If you feel nothing after 8 weeks, three possibilities: (1) peptide quality/purity issue — check your vendor COA; (2) dosing too low; (3) you're eating too close to your injections.

Q: Can I drink alcohol while on peptides?

A: Moderate alcohol (1-2 drinks) is generally fine for most peptides. However, alcohol acutely suppresses GH release for up to 24 hours. If you drink heavily the night of your injection, that GH pulse is largely wasted. For cognitive peptides (Semax, Selank) — alcohol is a CNS depressant and partially counteracts their effects. Avoid same-day alcohol use with any intranasal peptide.

Q: Can women use these peptides?

A: Yes. Most peptides work identically in men and women. Ipamorelin, CJC-1295, BPC-157, TB-500, Semax, Selank, TA-1, GHK-Cu, and GLP-1 agonists all have equivalent effects. PT-141 (Vyleesi) is actually FDA-approved specifically for women (HSDD). The main exception: GHRP-6 and other cortisol-elevating peptides may have more pronounced effects on cortisol in women — Ipamorelin's non-cortisol profile makes it even more preferred for women.

Q: What happens when I stop?

A: GH secretagogues: your IGF-1 gradually returns to baseline over 4-6 weeks. You keep the body composition gains IF you maintain training and diet. The pituitary does NOT stay suppressed — unlike exogenous HGH. GLP-1s: weight regain is common after stopping — typically 50-70% of weight lost returns within 1 year without lifestyle change. BPC-157/TB-500: healing effects are permanent once tissue is repaired.

Q: Can I use peptides while on TRT (testosterone replacement)?

A: Yes — GH peptides and TRT work on completely different axes. Ipamorelin + CJC-1295 + TRT is one of the most popular combinations for body recomposition. They are complementary, not competitive. BPC-157 is also commonly used with TRT for connective tissue support. No known dangerous interactions between standard TRT doses and research peptides.

Q: Why am I getting water retention? Is it bad?

A: Water retention in the hands, face, and ankles is extremely common with GH secretagogues — especially in the first 2-4 weeks. It's a sign the peptide is working. GH increases sodium and water reabsorption. It usually resolves after the first month as your body adapts. Reduce dose if severe. For MK-677, water retention can be more significant and persistent — this is a known side effect.

Q: How do I know if my peptide is fake/poor quality?

A: Key signs of underdosed or fake peptides: (1) No effect after 4+ weeks at proper dosing and timing; (2) Peptide dissolves instantly with no cloudiness — real peptides should create a slightly cloudy solution before fully dissolving; (3) No vendor-provided COA with HPLC purity above 98% and mass spectrometry confirmation. Always test a new vendor's product against a known one if possible. Finnrick.com has independent blind testing data on vendors.

Q: Is there a best time of year to run a peptide cycle?

A: For GH secretagogues: many users prefer to start cycles in fall/winter when natural GH levels are already slightly higher and caloric surplus for muscle gain is easier. Spring cycles work well for fat loss-focused protocols. Epitalon is traditionally run in spring and fall (2x/year). For healing peptides: start immediately after injury — do not wait. For GLP-1s: no seasonal preference — continuous use year-round.

12 Medical Disclaimer

Critical Legal & Safety Information

THIS IS NOT MEDICAL ADVICE

This guide is compiled for educational and research purposes only. The information does not constitute medical advice, diagnosis, or treatment recommendations. Many peptides described are research chemicals not approved for human consumption. Always consult a licensed physician before starting any peptide protocol.

Research Chemicals vs. FDA-Approved Drugs

The majority of peptides in this guide (Ipamorelin, CJC-1295, BPC-157, TB-500, Semax, Selank, etc.) are classified as research chemicals — they have NOT undergone the safety and efficacy trials required for human drug approval. FDA-approved exceptions include: Semaglutide, Tirzepatide, Liraglutide, Tesamorelin, Bremelanotide (PT-141/Vyleesi), Ziconotide, Leuprolide, Oxytocin, Octreotide, and Nesiritide.

Quality and Contamination Risk

Research peptides purchased online have no FDA quality oversight. They may be: mislabeled, underdosed, overdosed, contaminated with bacterial endotoxins, or completely different compounds. Only purchase from vendors providing batch-specific HPLC + mass spectrometry COAs from certified third-party laboratories.

2023 FDA Compounding Ban

As of November 2023, Ipamorelin, CJC-1295, GHRP-2, GHRP-6, and other peptides were removed from the FDA's 503A/503B bulk drug substances list. Licensed compounding pharmacies can no longer legally prepare these. They are available only as research chemicals from unregulated vendors.

Legal Status

Peptide laws vary by country and change frequently. In the US, possession of research chemicals is generally legal but selling them labeled for human use is not. In Australia, many peptides require a prescription. In the UK, most are Schedule 2 controlled substances. Always verify local laws before purchasing.

No Liability

The authors of this guide accept no liability for any harm arising from the use of information contained herein. The decision to use any compound described rests entirely with the individual and their physician. This guide does not encourage, facilitate, or endorse illegal activity.

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